Berkeley MSSE
MOLECULAR SCIENCE & SOFTWARE ENGINEERING

# Lecture 07:

## Correlation, Dimension Reduction and PCA

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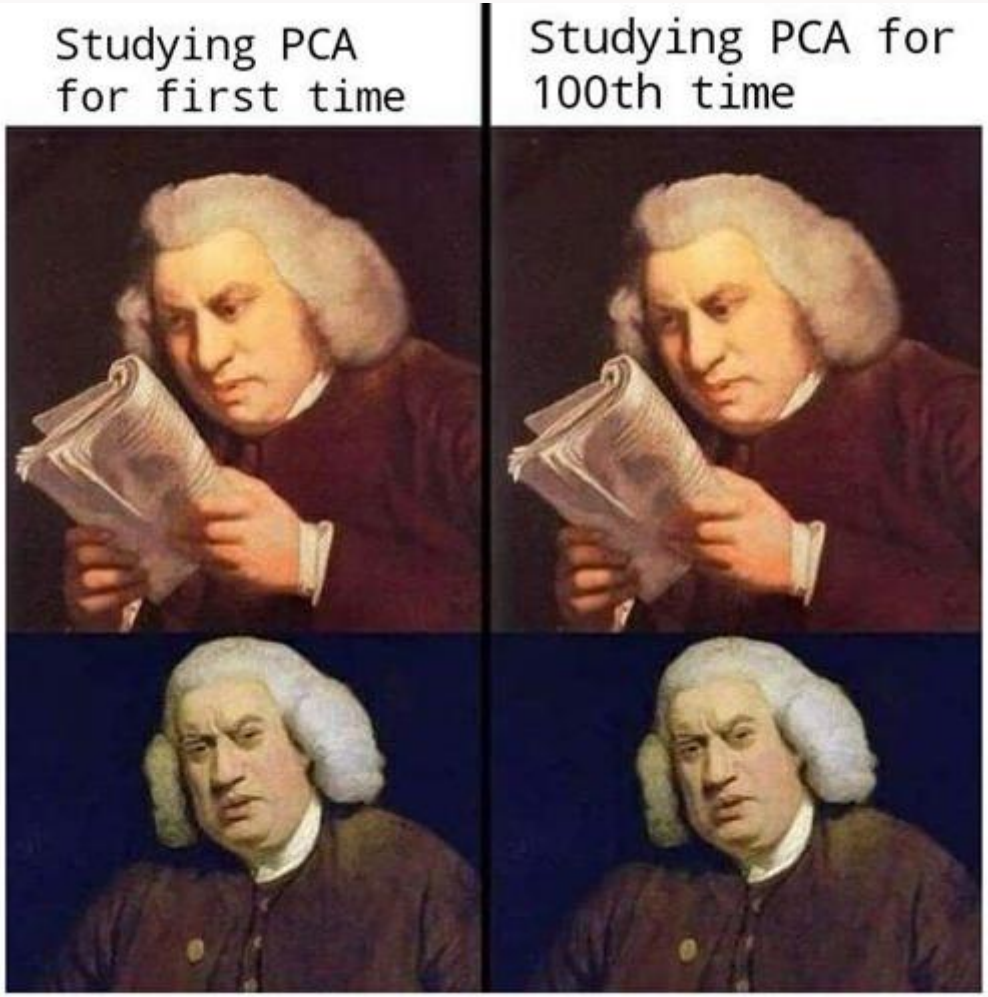
University California, Berkeley

**Data Science for Scientific Computing**

MSSE 277A, 3 Units

# Correlation, Dimension Reduction and PCA




credit: Raunak Joshi, @JackRaifer, X

<u>Outline</u>

**The curse of dimensionality**

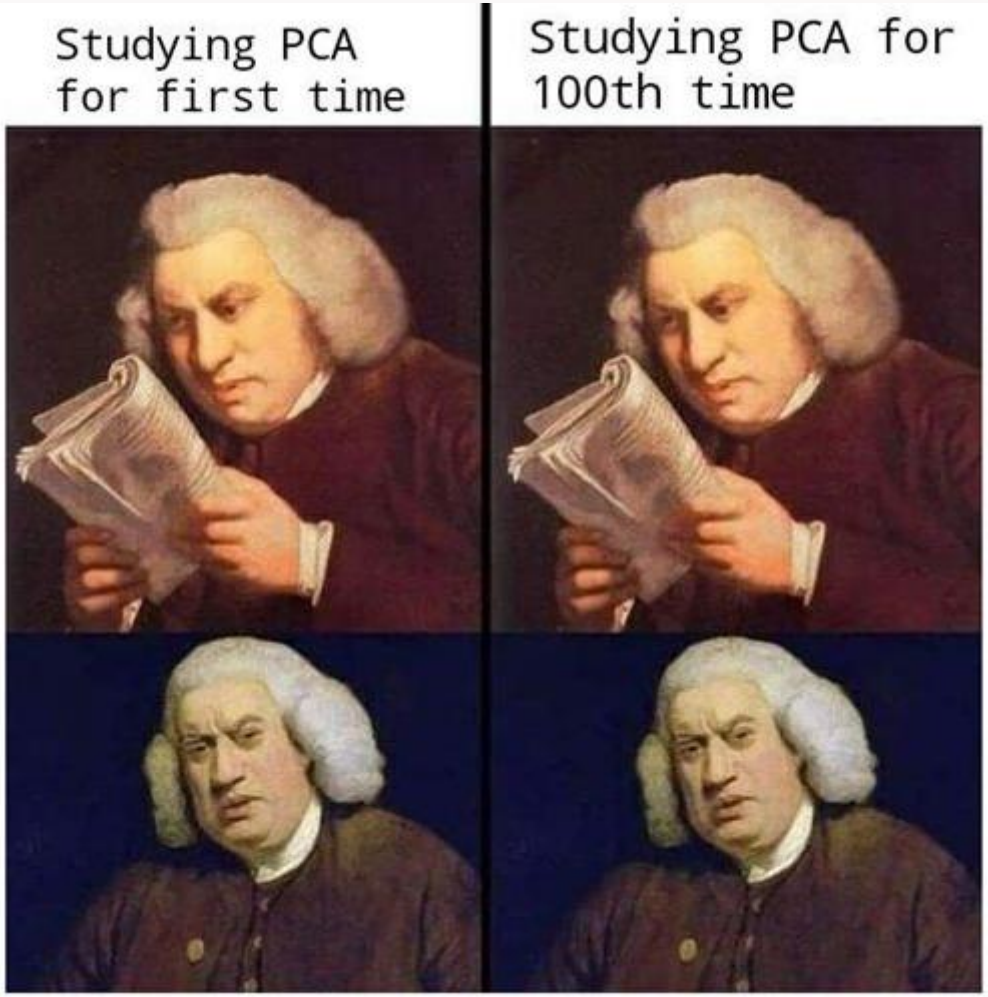
**Variance and Covariance**

**Correlation**

**Principal Component Analysis (PCA)**

**Alternatives**

# Correlation, Dimension Reduction and PCA

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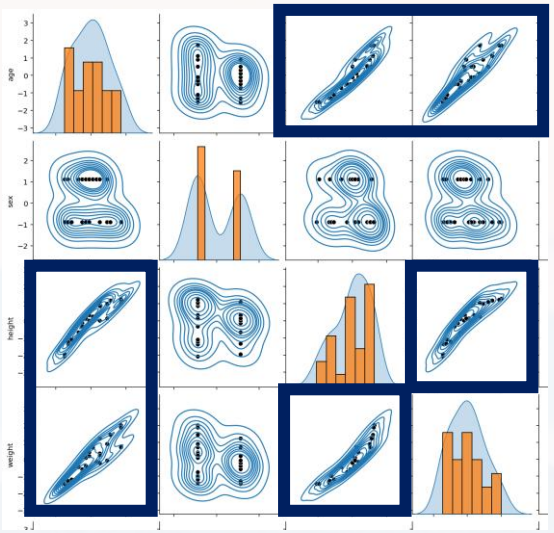
# Correlation, Dimension Reduction and PCA: The curse of dimensionality

| age | sex | height | weight | bmp | fev1 | rv | frc | tlc | pemax |
|---|---|---|---|---|---|---|---|---|---|
| 7 | 0 | 109 | 13.1 | 68 | 32 | 258 | 183 | 137 | 95 |
| 7 | 1 | 112 | 12.9 | 65 | 19 | 449 | 245 | 134 | 85 |
| 8 | 0 | 124 | 14.1 | 64 | 22 | 441 | 268 | 147 | 100 |

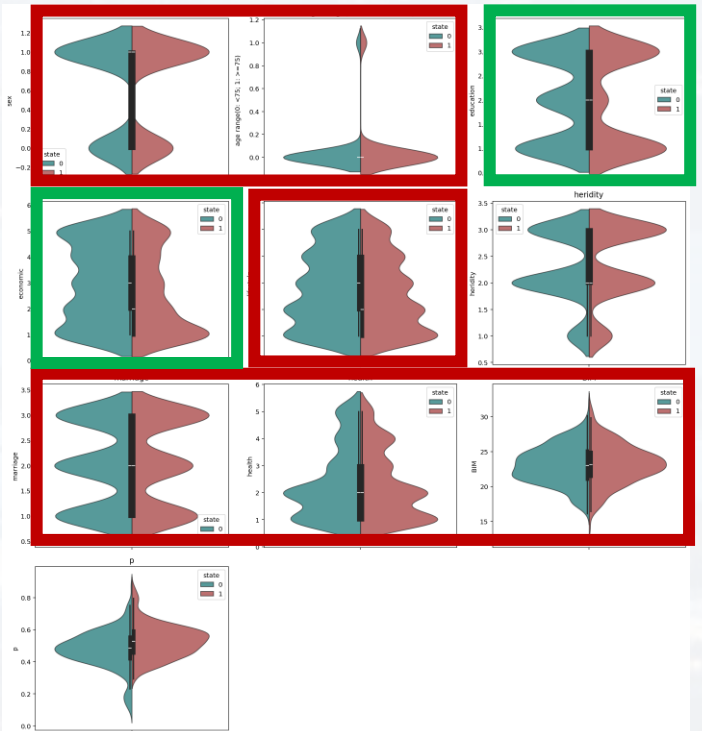


**some features are strongly related to each other → contain the same information**

| state | sex | ≥(0: <75; | education | economic | lifestyle | heridity | marriage | health | BIM | p |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 2 | 1 | 2 | 2 | 3 | 5 | 24.19 | 0.553777 |
| 0 | 0 | 1 | 1 | 4 | 1 | 1 | 3 | 5 | 18.05 | 0.375283 |
| 0 | 0 | 0 | 3 | 2 | 2 | 3 | 3 | 5 | 21.98 | 0.541784 |

run `ViolinePlotStatesAD.py`

states: 1 = AD
0 = healthy



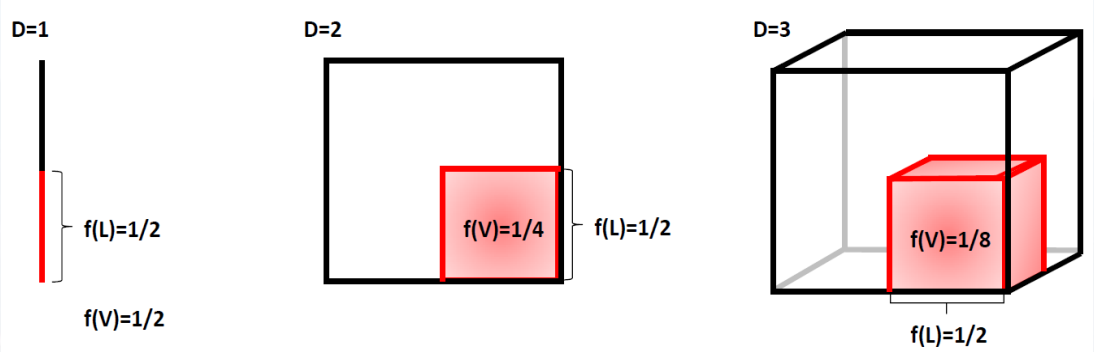
**some features don't differ between classes → not helpful for classification**

**some features do differ between classes → needed for classification**

# Correlation, Dimension Reduction and PCA: The curse of dimensionality

We don't need all the features. Some are **more relevant** than others and some are **more related** than others.
→ **reducing** the number of features, i. e. **dimensionality**
But why should we want to do that?

*What is the fraction of volume $f(V)$ covered by a certain fraction of length $f(L) = r/R$ for different dimensions D?*




**answer:** $\boldsymbol{f(V) \sim f(L)^D}$

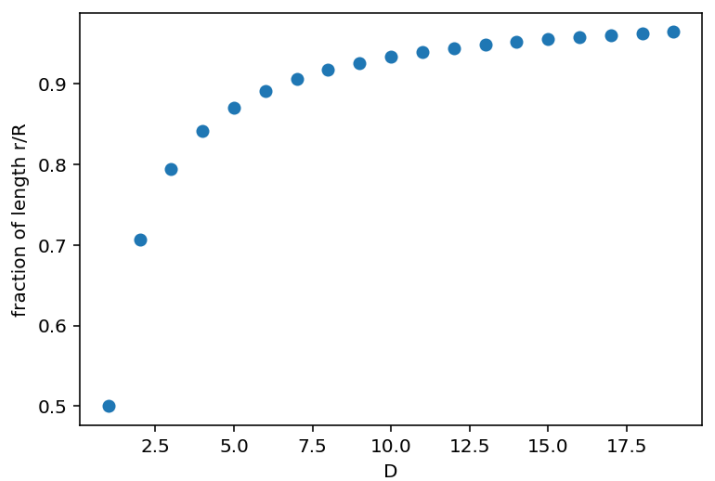
$$f(V) = C(D) f(L)^D$$

$$= C(D) \left(\frac{r}{R}\right)^D$$

$C(D)$: constant that only depends on D

$$\left(\frac{r}{R}\right) \sim \sqrt[D]{f(V)}$$

fraction of length that we need to explore in order to find a particular fraction of datapoints **(uniformly distributed in V)**



# Correlation, Dimension Reduction and PCA: The curse of dimensionality

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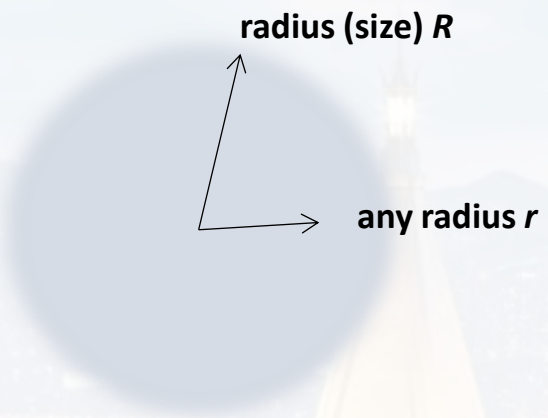
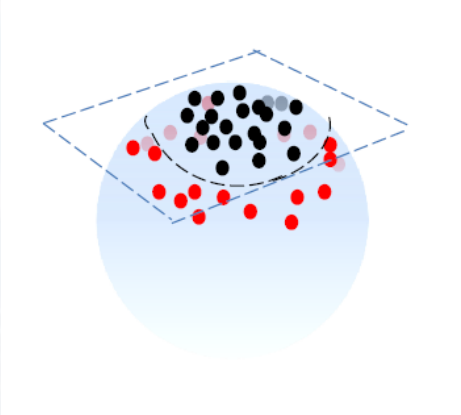
N - D space



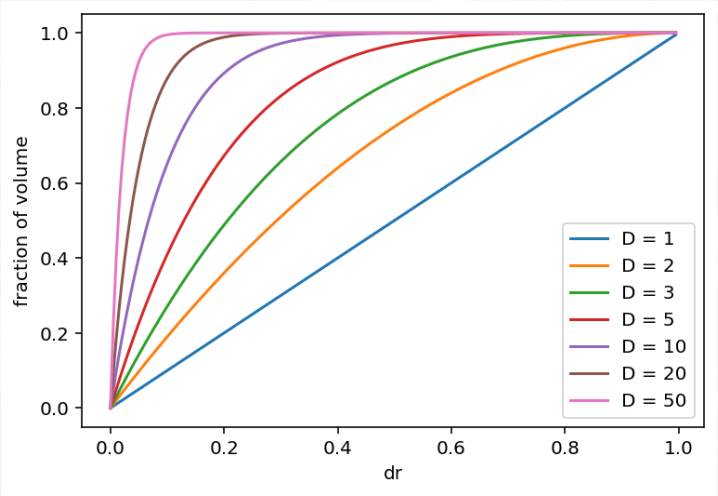
hypersphere:

$C(D)$: constant that only depends on D

$$V_D(r) = C(D)\, r^D$$

fraction of volume between $r$ and $r - dr$

$$\frac{V_D(r) - V_D(r - dr)}{V_D(r)} = 1 - \frac{(r - dr)^D}{r^D}$$




Most of the volume of a hypersphere is located on its surface!

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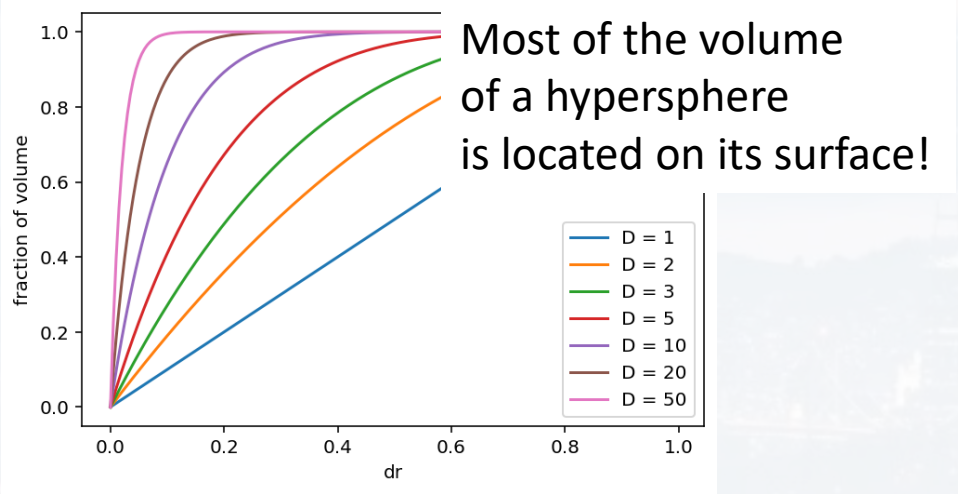


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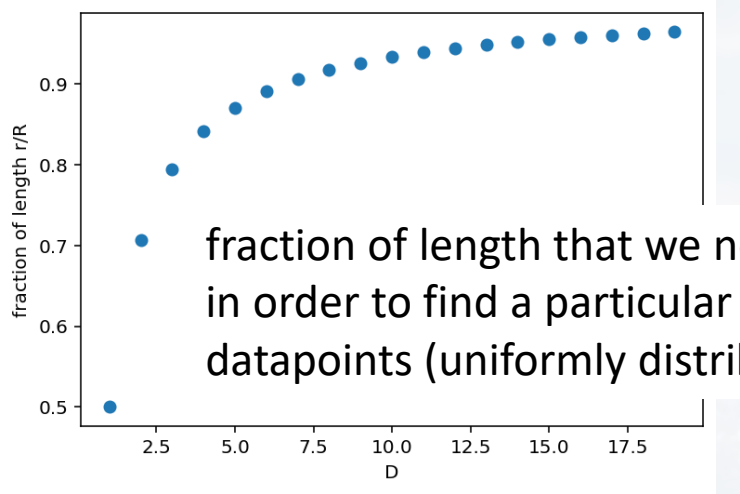


fraction of length that we need to explore in order to find a particular fraction of datapoints (uniformly distributed in R)

- for **large D (= number of dimensions)**, one has to explore a **larger fraction $\frac{r}{R}$ of the data space** in order to get the **same fraction of data points**
- many algorithms get **less efficient** for large D
- for D→∞, the entire volume is located on the surface of the hyperspace

# Correlation, Dimension Reduction and PCA: The curse of dimensionality

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- for **large D (= number of dimensions)**, one has to explore a **larger fraction $\frac{r}{R}$ of the data space** in order to get the **same fraction of data points**
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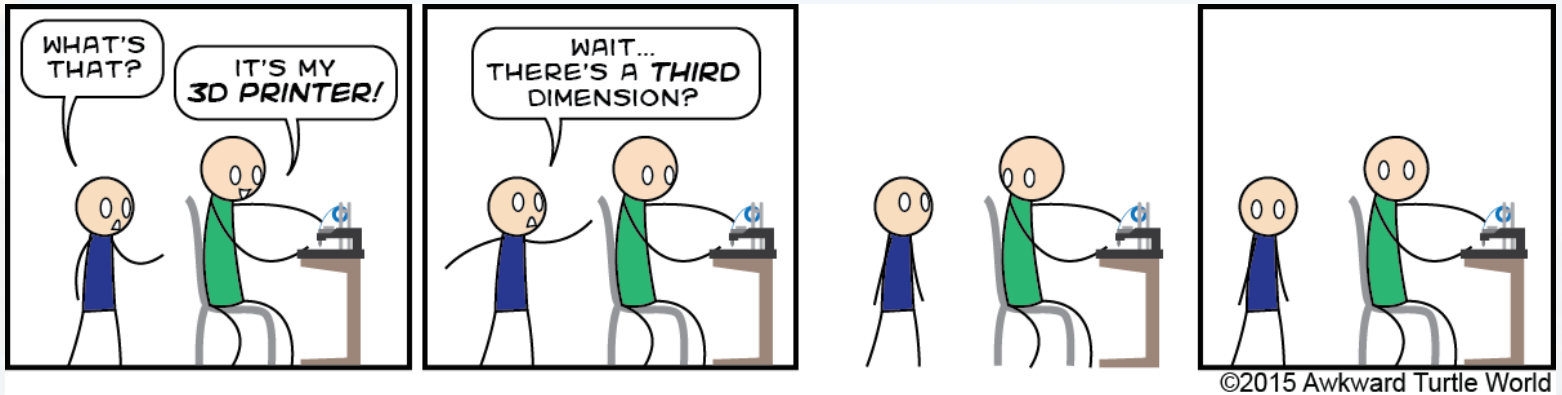
- for D→∞, the entire volume is located on the surface of the hyperspace

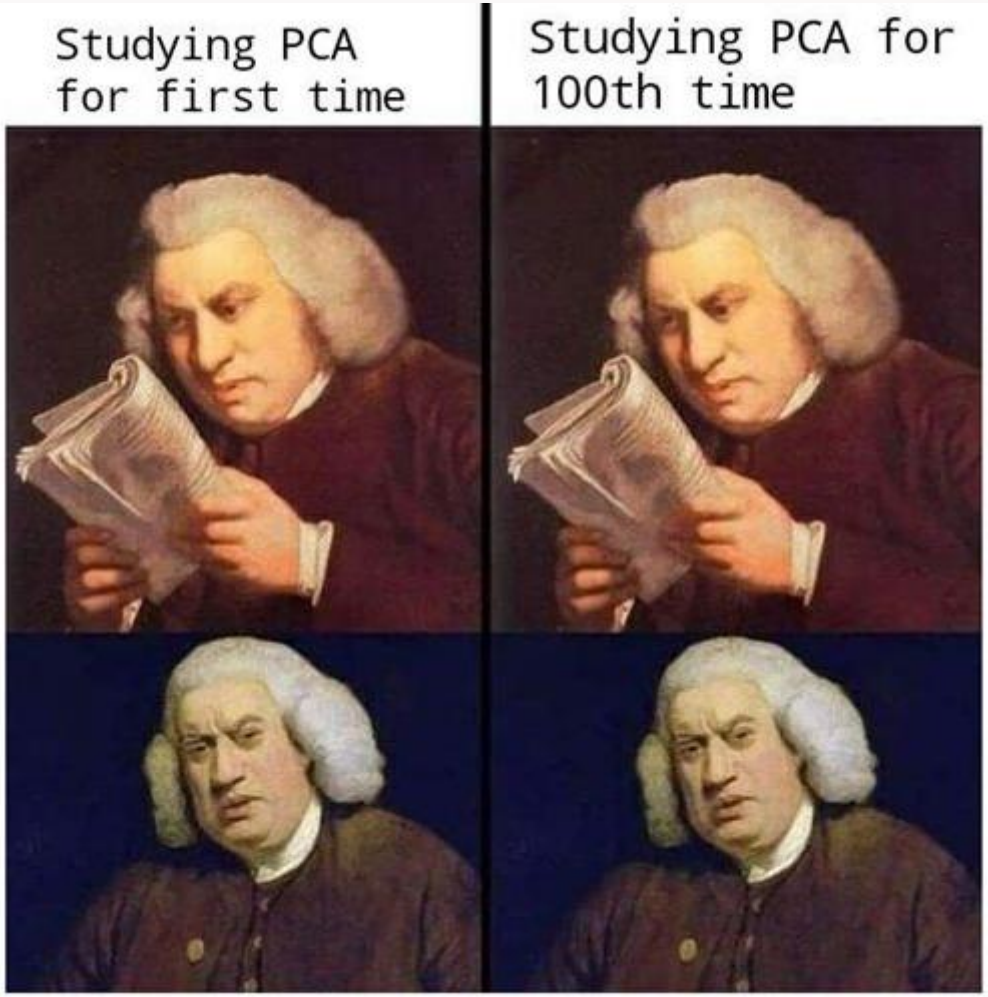
Makes many machine learning algorithms less efficient!

**"Curse of Dimensionality"**

**Dimensionality reduction is important!**



# Correlation, Dimension Reduction and PCA

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The curse of dimensionality

**Variance and Covariance**

Correlation

Principal Component Analysis (PCA)

Alternatives

| | the mean $\mu$ | the variance $\sigma^2$ |
|---|---|---|
| | (barycenter) | (natural scatter) |
| discrete (= countable) | $\mu = E(x) = \sum_i x_i \, p(x_i)$ | $\sigma^2 = var(x) = \sum_i (x_i - \mu)^2 \, p(x_i)$ |
| continuous | $\mu = E(x) = \int x \, p(x) \, dx$ | $\sigma^2 = var(x) = \int (x - \mu)^2 \, p(x) \, dx$ |

# Correlation, Dimension Reduction and PCA: Variance and Covariance

$$\mu = E(x) = \int x\, p(x)\, dx$$

$$\sigma^2 = var(x) = \int (x-\mu)^2\, p(x)\, dx$$

$$var(x) = \int (x-\mu)^2\, p(x)\, dx = E([x-\mu]^2)$$

variance can be interpreted as **mean of $[x-\mu]^2$**

$$= E(x^2 - 2x\mu + \mu^2)$$

$$= \int [x^2 - 2x\mu + \mu^2]\, p(x)\, dx$$

$$= \int x^2\, p(x)\, dx - 2\mu \int x\, p(x)\, dx + \mu^2 \int p(x)\, dx$$

$$= E(x^2) - 2\mu\, E(x) + \mu^2\, E(1)$$

**2nd axiom** $\int p(x)\, dx = 1$

$$= E(x^2) - 2\mu\, E(x) + \mu^2$$

$\mu = E(x)$

$$\boldsymbol{\sigma^2 = E(x^2) - E(x)^2}$$

# Correlation, Dimension Reduction and PCA: Variance and Covariance

$$\boxed{\sigma^2 = E(x^2) - E(x)^2}$$

$$\mu = E(x) = \int x\, p(x)\, dx$$

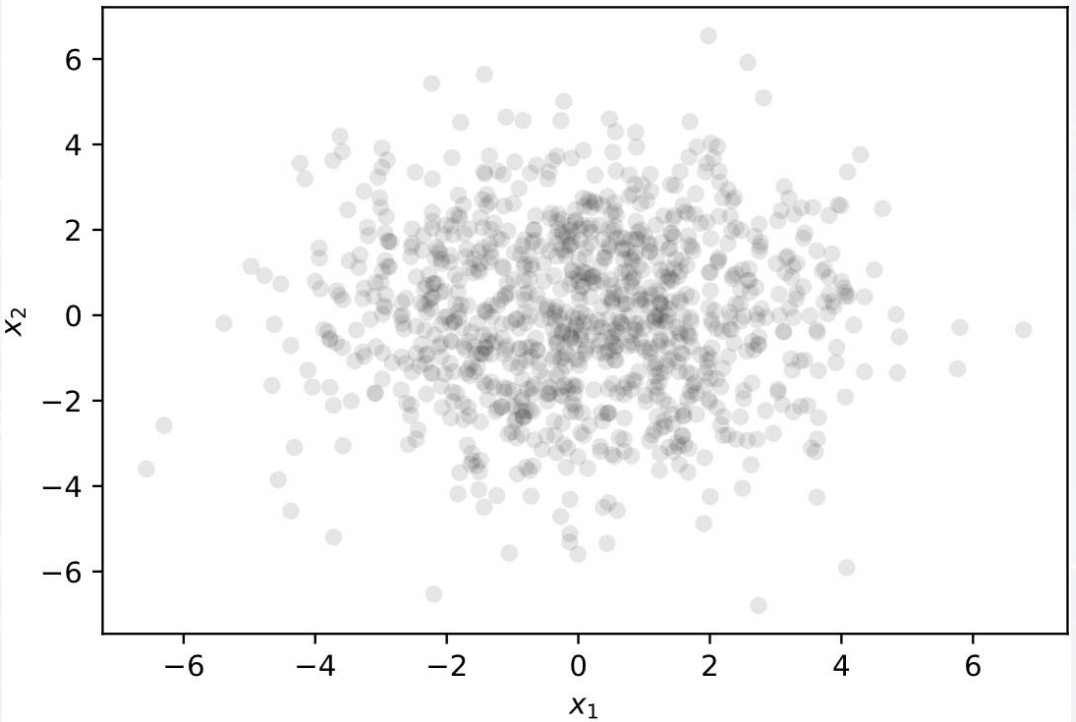
$$\sigma^2 = var(x) = \int (x-\mu)^2\, p(x)\, dx$$

plotting two sets of random number: $x_1$ and $x_2$

```
x1 = np.random.normal(0,2,(1000,))
x2 = np.random.normal(0,2,(1000,))

plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')
plt.xlabel('$x_1$')
plt.ylabel('$x_2$')
```

$x_1$ and $x_2$ are unrelated and mutually **independent**
→ featureless data cloud



# Correlation, Dimension Reduction and PCA: Variance and Covariance

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```
x1 = np.random.normal(0,2,(1000,))
x2 = np.random.normal(0,20,(1000,))

plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')
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```

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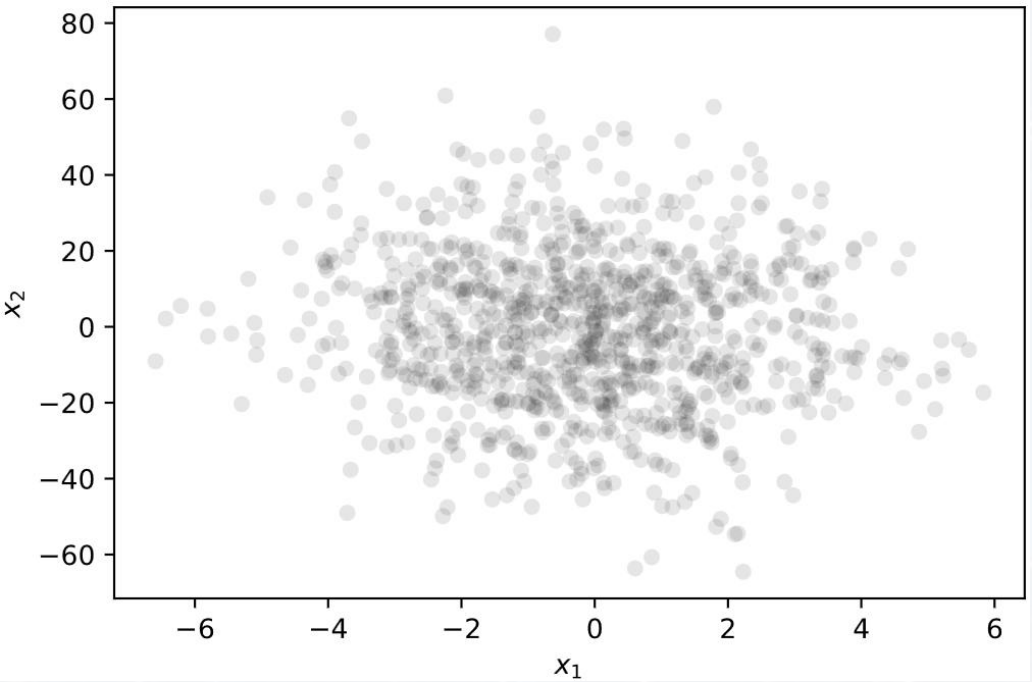

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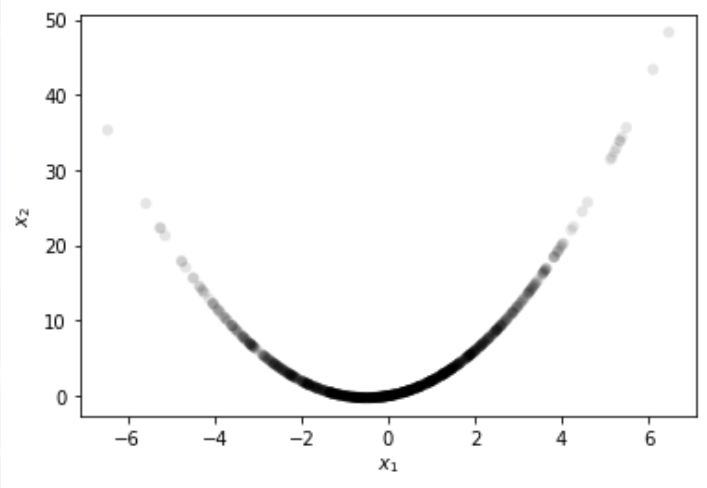
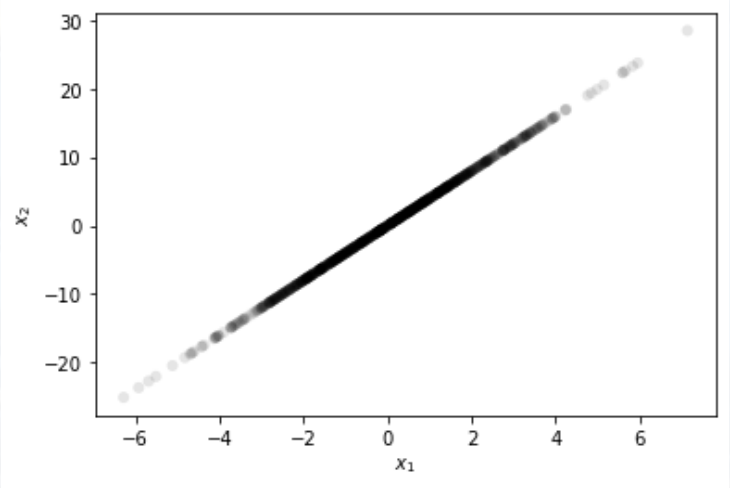
plotting two sets of random number: $x_1$ and $x_2$

```
x1 = np.random.normal(0,2,(1000,))
x2 = x1**2 + x1
#x2 = 4*x1

plt.scatter(x1, x2, color = 'k', alpha = 0.1, edgecolor = 'none')
plt.xlabel('$x_1$')
plt.ylabel('$x_2$')
```

based on the shape of the data cloud
→ prediction how $x_1$ and $x_2$ are related, i. e. how they **correlate**



# Correlation, Dimension Reduction and PCA: Variance and Covariance

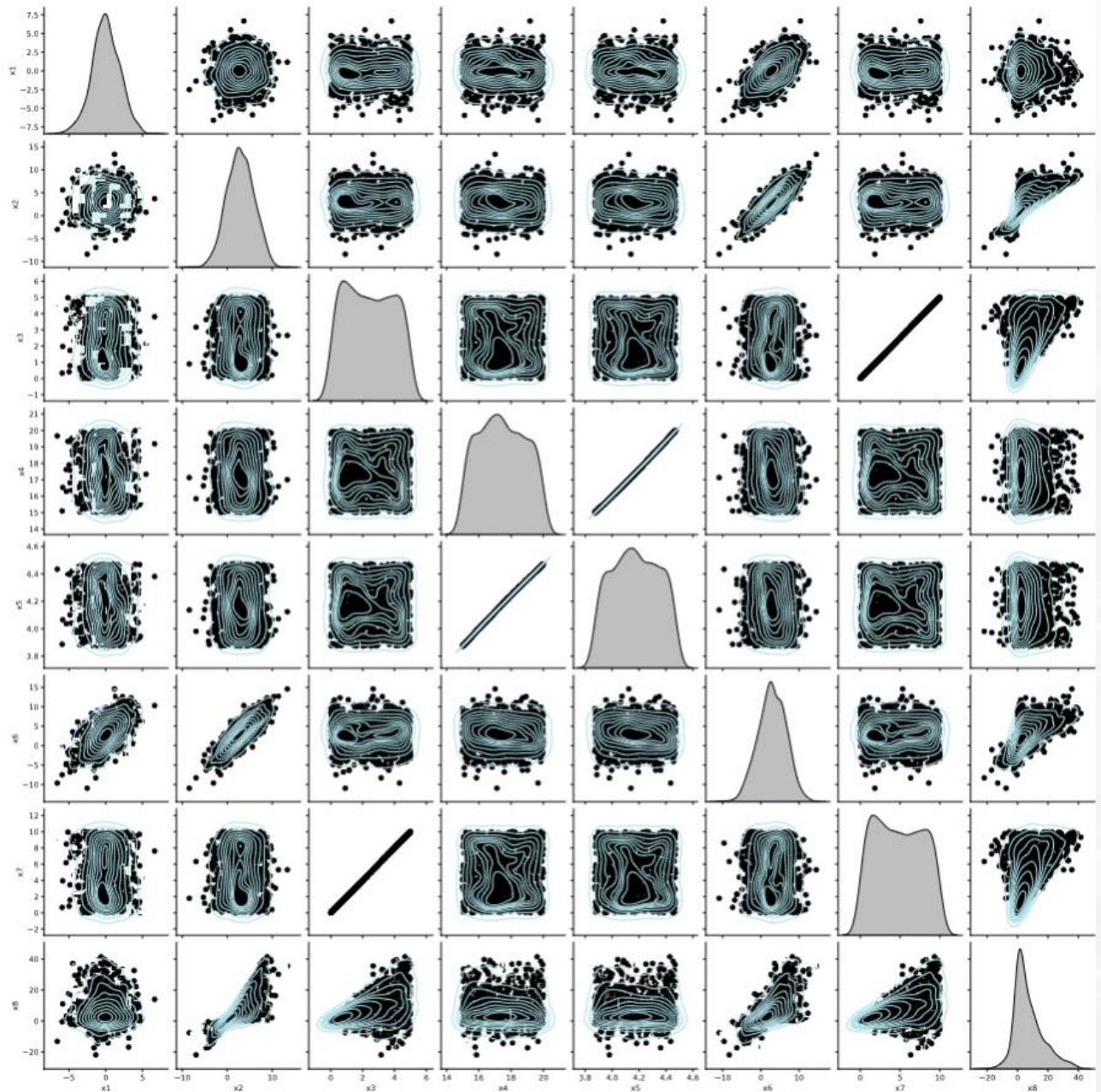
$$\boxed{\sigma^2 = E(x^2) - E(x)^2}$$

$$\mu = E(x) = \int x\, p(x)\, dx$$

$$\sigma^2 = var(x) = \int (x-\mu)^2\, p(x)\, dx$$

```
x1 = np.random.normal(0,2,(1000,))
x2 = np.random.normal(3,3,(1000,))

x3 = np.random.uniform(0,5,(1000,))
x4 = 5*np.random.uniform(3,4,(1000,))

x5 = np.sqrt(x4)
x6 = x1 + x2
x7 = 2*x3
x8 = x3*x2

All  = np.vstack((x1, x2, x3, x4, x5, x6, x7, x8))
data = pd.DataFrame(All.transpose(),
                    columns = ['x1', 'x2', 'x3', 'x4', 'x5', 'x6', 'x7', 'x8'])

out = sns.pairplot(data, kind = "kde", \
                   plot_kws = {'color':[176/255, 224/255, 230/255]}, \
                   diag_kws = {'color':'black'})
out.map_offdiag(plt.scatter, color = 'black')
```

# Correlation, Dimension Reduction and PCA: Variance and Covariance



$$\mu = E(x) = \int x\, p(x)\, dx$$

$$\sigma^2 = var(x) = \int (x-\mu)^2\, p(x)\, dx$$

based on the shape of the data cloud

→ prediction how $x_1$ and $x_2$ are related, i. e. how they **correlate**

→ how to quantify?

# Correlation, Dimension Reduction and PCA: Variance and Covariance

a, b = const

$$\boxed{\sigma^2 = E(x^2) - E(x)^2}$$

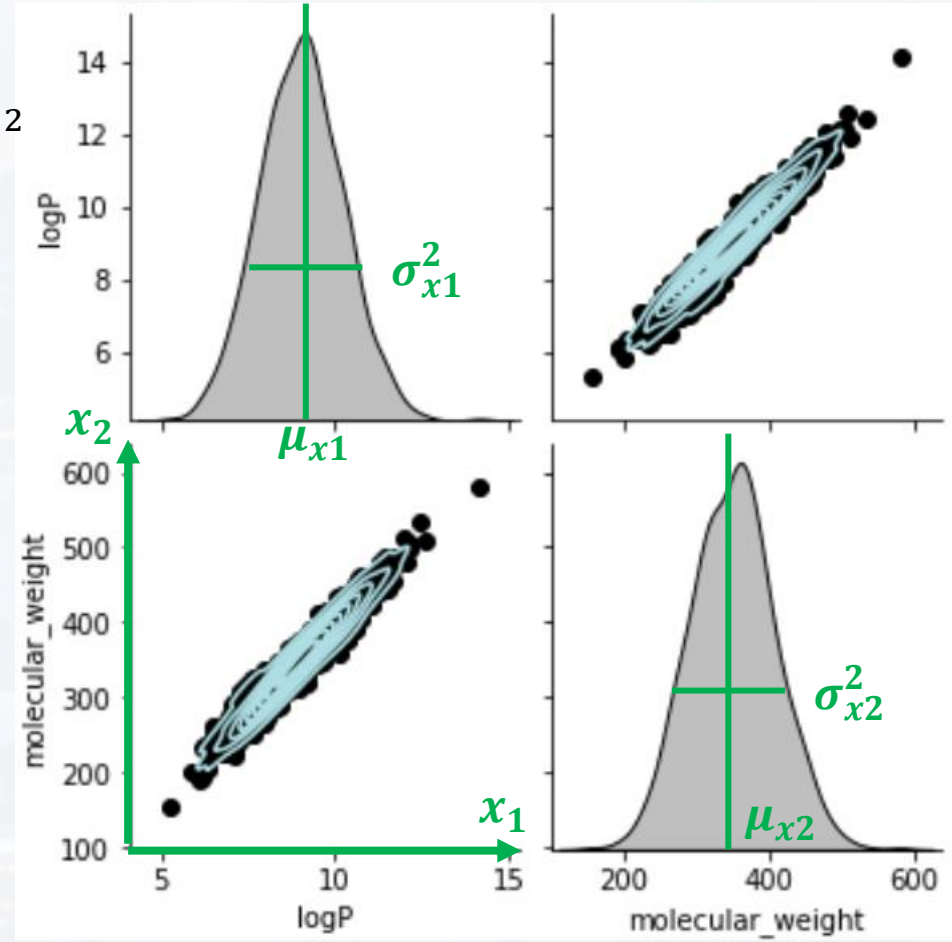
$$\mu = E(x) = \int x\, p(x)\, dx$$

$$\sigma^2 = var(x) = \int (x-\mu)^2\, p(x)\, dx$$

$$var([a\, x_1 + b\, x_2]) = E([a\, x_1 + b\, x_2]^2) - E(a\, x_1 + b\, x_2)^2$$

$$= E(a^2\, x_1^2 + 2ab\, x_1 x_2 + b^2\, x_2^2) - E(a\, x_1 + b\, x_2)^2$$

$$= a^2 E(x_1^2) + 2ab\, E(x_1 x_2) + b^2\, E(x_2^2) - E(a\, x_1 + b\, x_2)^2$$



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$$= E(a^2\, x_1^2 + 2ab\, x_1 x_2 + b^2\, x_2^2) - E(a\, x_1 + b\, x_2)^2$$

$$= a^2 E(x_1^2) + 2ab\, E(x_1 x_2) + b^2\, E(x_2^2) - E(a\, x_1 + b\, x_2)^2$$

$$= a^2 E(x_1^2) + 2ab\, E(x_1 x_2) + b^2\, E(x_2^2) - [aE(x_1) + b\, E(x_2)]^2$$

$$= \underbrace{a^2 E(x_1^2) - a^2 E(x_1)^2}_{a^2\, var(x_1)} + \underbrace{b^2\, E(x_2^2) - b^2 E(x_2)^2}_{b^2\, var(x_2)} + \underbrace{2ab\, E(x_1 x_2) - 2abE(x_1)E(x_2)}_{2ab\, cov(x_1, x_2)}$$

$$= a^2\, var(x_1) + b^2\, var(x_2) + 2ab\, cov(x_1, x_2)$$

$$\boxed{cov(x_1, x_2) = cov(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)}$$

**covariance**

# Correlation, Dimension Reduction and PCA: Variance and Covariance

$$\boldsymbol{cov(x_1, x_2) = cov(x_2, x_1)} = E(x_1 x_2) - E(x_1)E(x_2)$$

$$\mu = E(x) = \int x\, p(x)\, dx$$

$$\sigma^2 = var(x) = \int (x-\mu)^2\, p(x)\, dx$$

What does the covariance tell us?

1) geometrical interpretation → PCA
2) arithmetical interpretation

# Correlation, Dimension Reduction and PCA: Variance and Covariance

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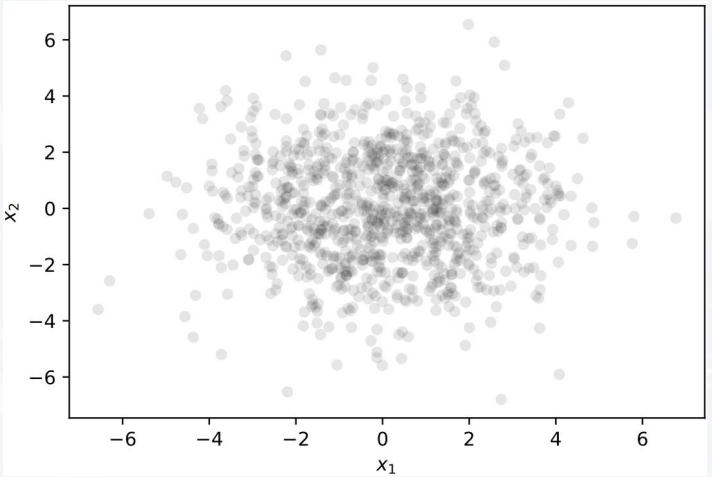


a) $x_1$ and $x_2$ are independent

$$E(x_1 x_2) - E(x_1)E(x_2)$$

$$= \iint x_1 x_2\, p(x_1)\, p(x_2)\, dx_1 dx_2 - \int x_1\, p(x_1)\, dx_1 \int x_2\, p(x_2)\, dx_2$$

$x_1$ and $x_2$ are independent:

$x_1$ is not a function of $x_2$ and vise verse
$x_1$ cannot be predicted by $x_2$ and vise verse

$$= \int x_1\, p(x_1)\, dx_1 \int x_2\, p(x_2)\, dx_2 - \int x_1\, p(x_1)\, dx_1 \int x_2\, p(x_2)\, dx_2 = 0$$

Covariance equals **zero** if samples are **independent**!

# Correlation, Dimension Reduction and PCA: Variance and Covariance

$$\boldsymbol{cov(x_1, x_2) = cov(x_2, x_1)} = E(x_1 x_2) - E(x_1)E(x_2)$$

$$\mu = E(x) = \int x\, p(x)\, dx$$

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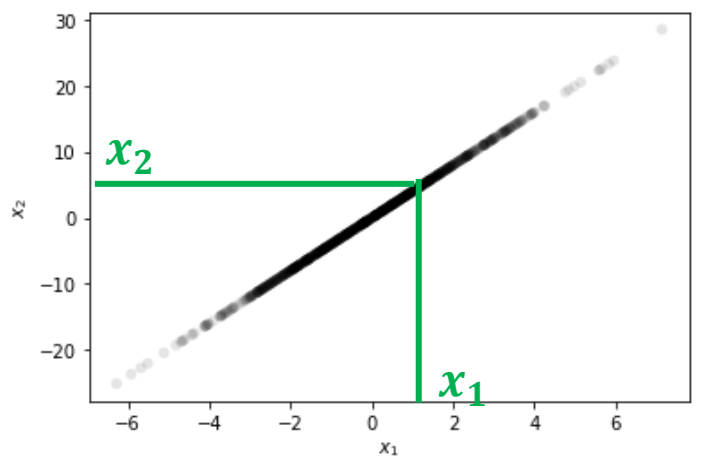


b) $x_1$ and $x_2$ are **not** independent

$$E(x_1 x_2) - E(x_1)E(x_2)$$

$$= \iint x_1 x_2\, p(x_1)\, p(x_2)\, dx_1 dx_2 - \int x_1\, p(x_1)\, dx_1 \int x_2\, p(x_2)\, dx_2$$

$x_1$ and $x_2$ are **not** independent:

$x_1$ ***is*** a function of $x_2$ and vise verse
$x_1$ ***can*** be predicted by $x_2$ to certain degree and vise verse

$$= \iint x_1\, p(x_1)\;\; x_2(x_1)\, p(x_2(x_1))\, dx_1 dx_2(x_1) - \int x_1\, p(x_1)\, dx_1 \int x_2\, p(x_2)\, dx_2$$

Covariance **does not** equal **zero**!

# Correlation, Dimension Reduction and PCA: Variance and Covariance

covariance matrix

$$\Sigma = \begin{pmatrix} cov(x_1, x_1) & \dots cov(x_1, x_j) \dots & cov(x_1, x_I) \\ \dots & \dots & \dots \\ cov(x_i, x_1) & \dots cov(x_i, x_i) \dots & cov(x_i, x_I) \\ \dots & \dots & \dots \\ cov(x_I, x_1) & \dots cov(x_I, x_j) \dots & cov(x_I, x_I) \end{pmatrix}$$

$$cov(x_i, x_j) = E(x_i x_j) - E(x_i)E(x_j)$$

for $i = j$

$$cov(x_i, x_i) = E(x_i x_i) - E(x_i)E(x_i)$$

$$= E(x_i^2) - E(x_i)^2$$

$$= \sigma^2(x_i)$$

# Correlation, Dimension Reduction and PCA: Variance and Covariance

covariance matrix

$$\Sigma = \begin{pmatrix} \boldsymbol{cov(x_1, x_1)} & \dots cov(x_1, x_j) \dots & cov(x_1, x_I) \\ \dots & \dots & \dots \\ cov(x_i, x_1) & \dots \boldsymbol{cov(x_i, x_i)} \dots & cov(x_i, x_I) \\ \dots & \dots & \dots \\ cov(x_I, x_1) & \dots cov(x_I, x_j) \dots & \boldsymbol{cov(x_I, x_I)} \end{pmatrix}$$

**diagonal = variance of the variable**

$$cov(x_i, x_j) = E(x_i x_j) - E(x_i)E(x_j)$$

for $i = j$

$$cov(x_i, x_i) = E(x_i x_i) - E(x_i)E(x_i)$$

$$= E(x_i^2) - E(x_i)^2$$

$$= \sigma^2(x_i)$$

# Correlation, Dimension Reduction and PCA: Variance and Covariance

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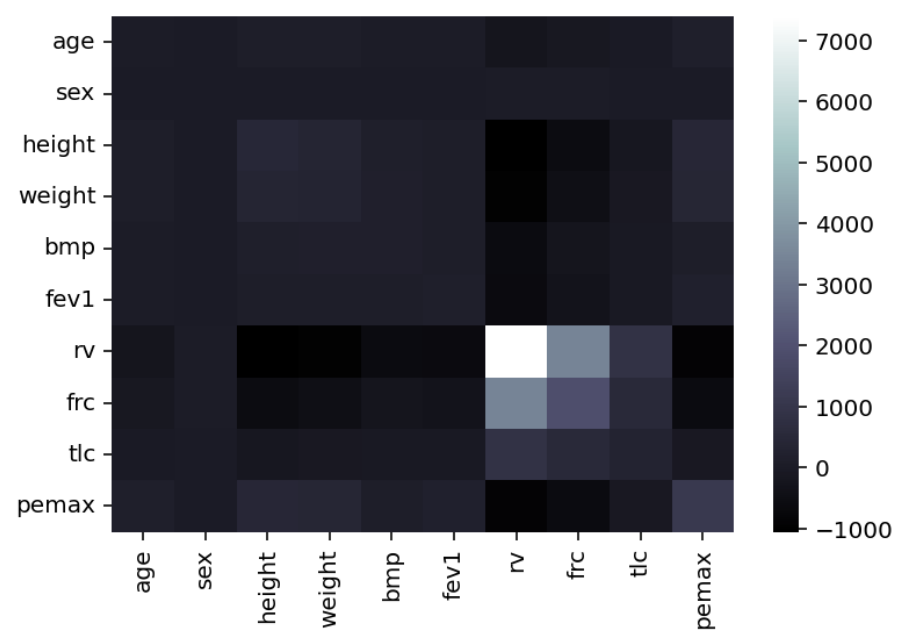
**diagonal = variance of the variable**

```
data      = pd.read_csv('../Datasets/Cystfibr.txt', delimiter = '\t')
sns.heatmap(data.cov(), cmap = 'bone' )
```




**note:** covariance is **not** scale invariant!

$$cov(x_i, x_j) = \iint x_i, x_j \, p(x_i) \, p(x_j) \, dx_i dx_j$$
$$- \int x_i \, p(x_j) \, dx_i \int x_i \, p(x_j) \, dx_j$$

# Correlation, Dimension Reduction and PCA: Variance and Covariance

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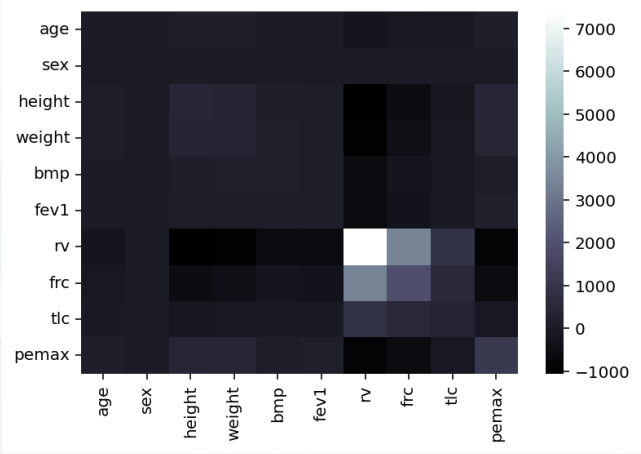
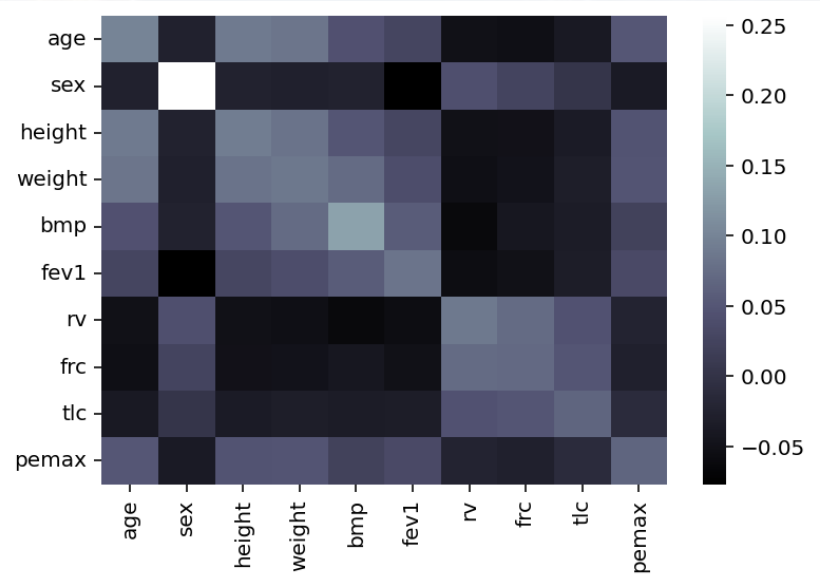
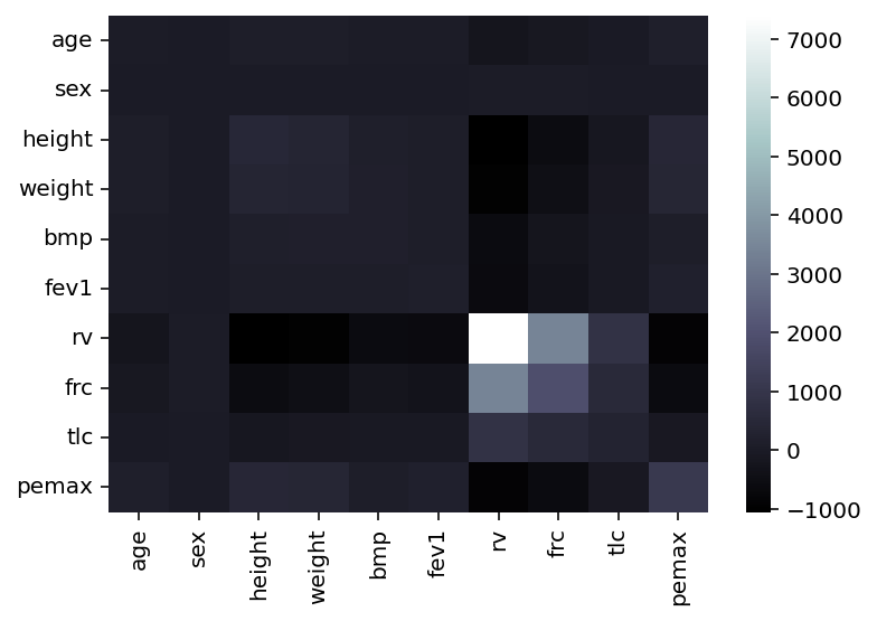



```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

data_scaled = scaler.fit_transform(data)
data_scaled = pd.DataFrame(data_scaled, columns = data.columns)

sns.heatmap(data_scaled.cov(), cmap = 'bone')
```



# Correlation, Dimension Reduction and PCA: Variance and Covariance

**note:** covariance is **not** scale invariant!

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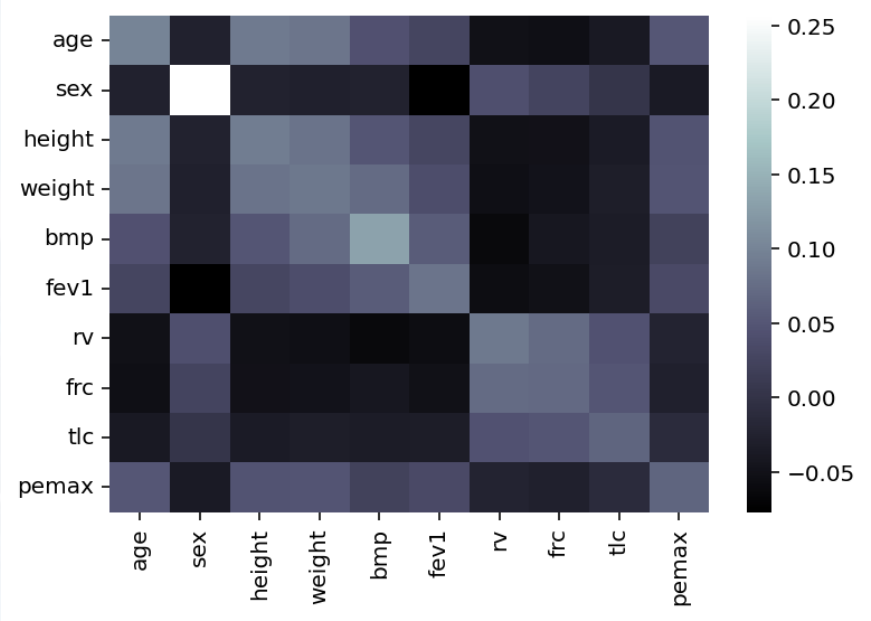


```
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```
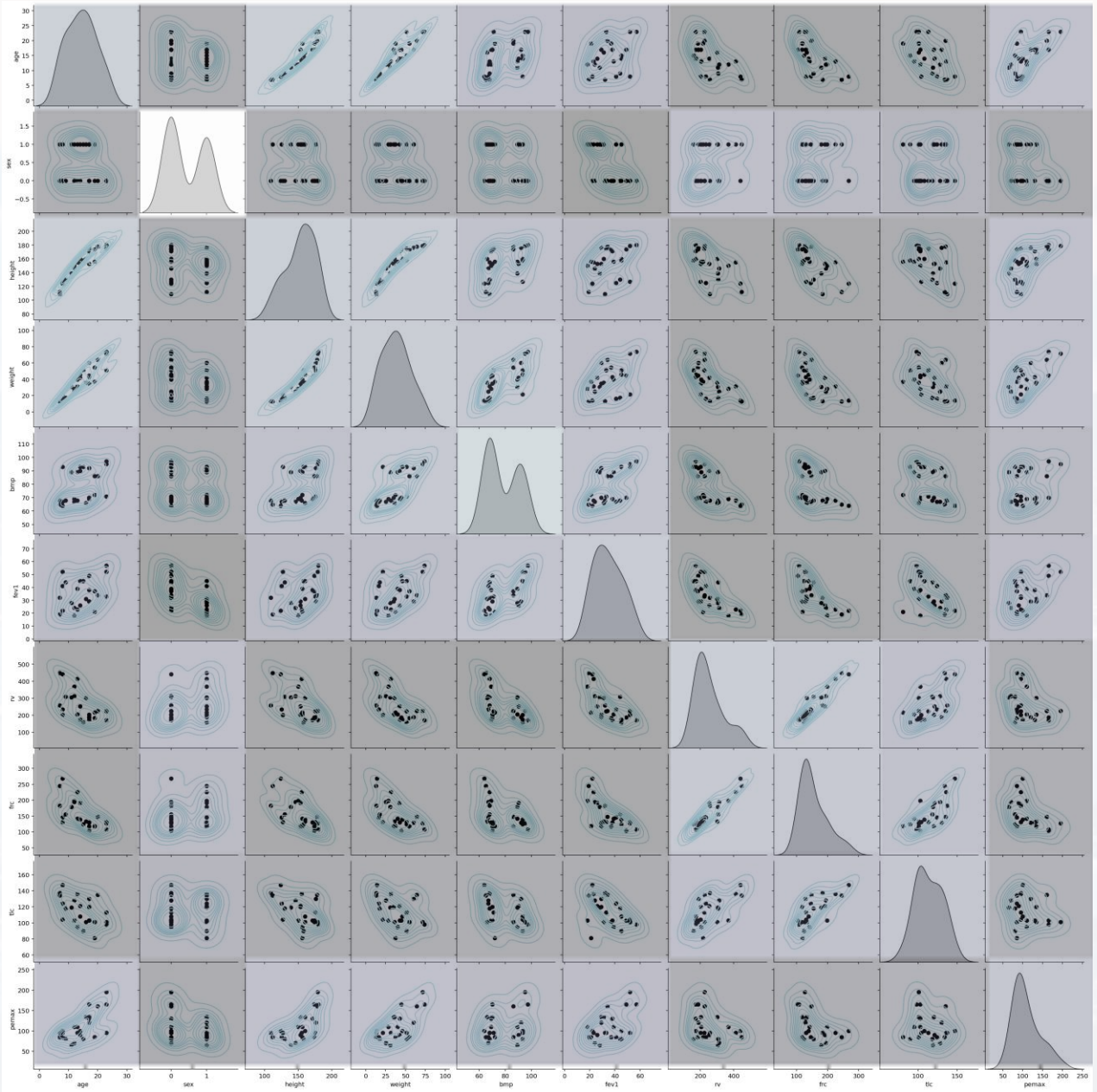
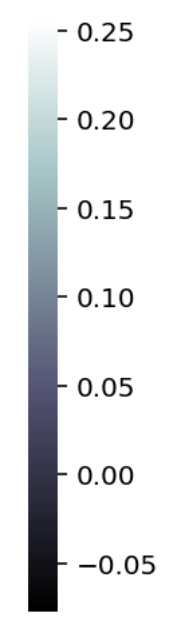
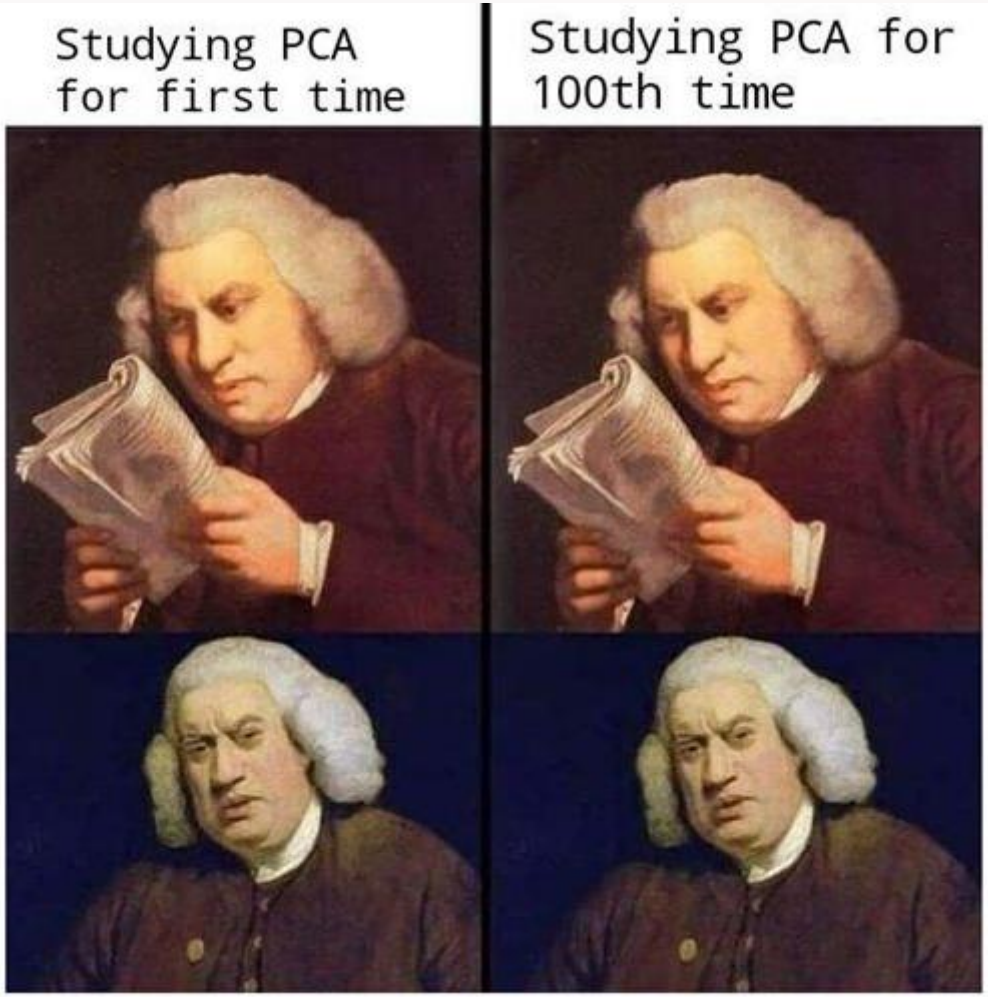

# Correlation, Dimension Reduction and PCA: Variance and Covariance




covariance gives us some idea on how the different features relate to each other

# Correlation, Dimension Reduction and PCA




credit: Raunak Joshi, @JackRaifer, X

Outline

The curse of dimensionality

Variance and Covariance

**Correlation**

Principal Component Analysis (PCA)

Alternatives

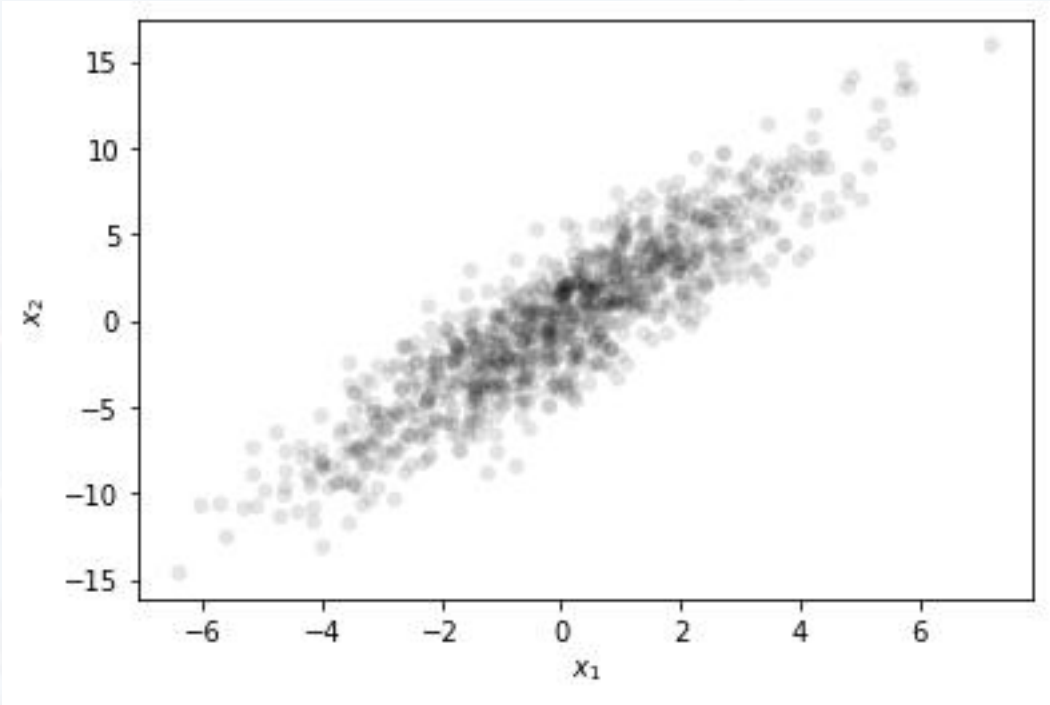
# Correlation, Dimension Reduction and PCA:



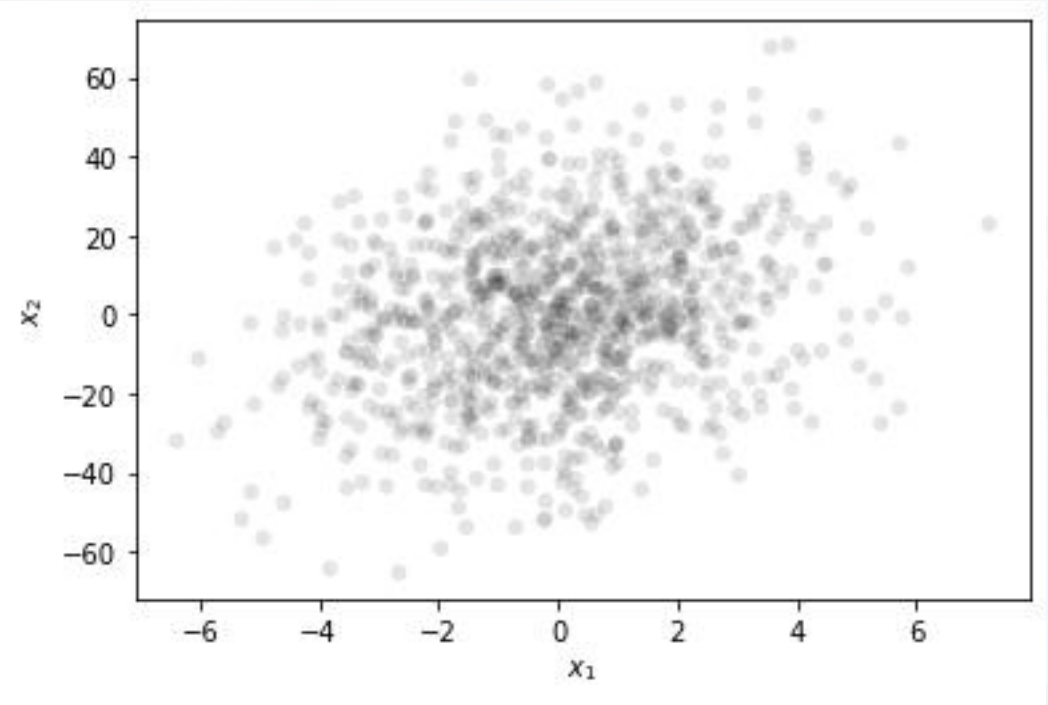
$$\boxed{cov(x_1, x_2) = cov(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)}$$

**covariance**

```
x1 = np.random.normal(0, 2, (1000,))
x2 = 2*x1 + np.random.normal(0, 2, (1000,))
```




```
x1 = np.random.normal(0, 2, (1000,))
x2 = 2*x1 + np.random.normal(0, 20, (1000,))
```



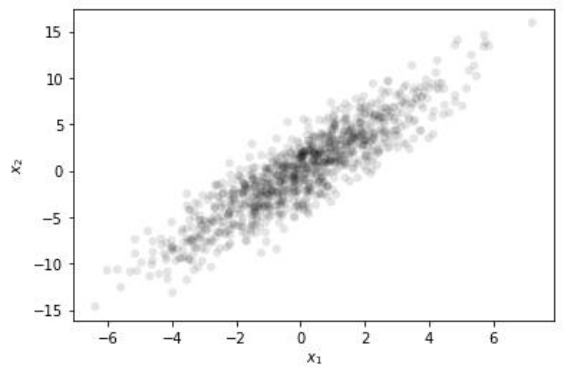

Same dependency, but different variance!

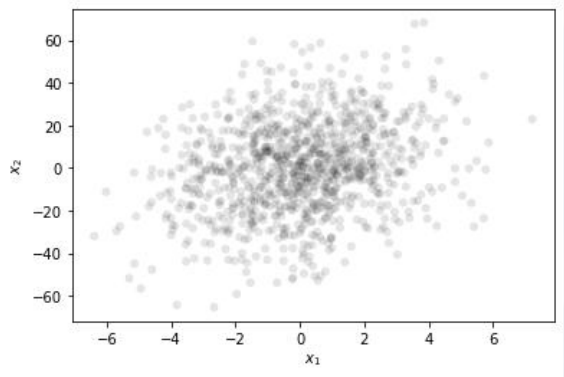
# Correlation, Dimension Reduction and PCA:



$$cov(x_1, x_2) = cov(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

**covariance**




Same dependency, but different variance!

**Need to scale for the variance!**

**Pearson's correlation coefficient**

$$\rho(x_1, x_2) = \frac{cov(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

$\rho(x_1, x_2)$:

- ranges from -1 to +1
- zero: no correlation **(completely independent)**
- -1: max anti correlation
- +1: max correlation

# Correlation, Dimension Reduction and PCA:



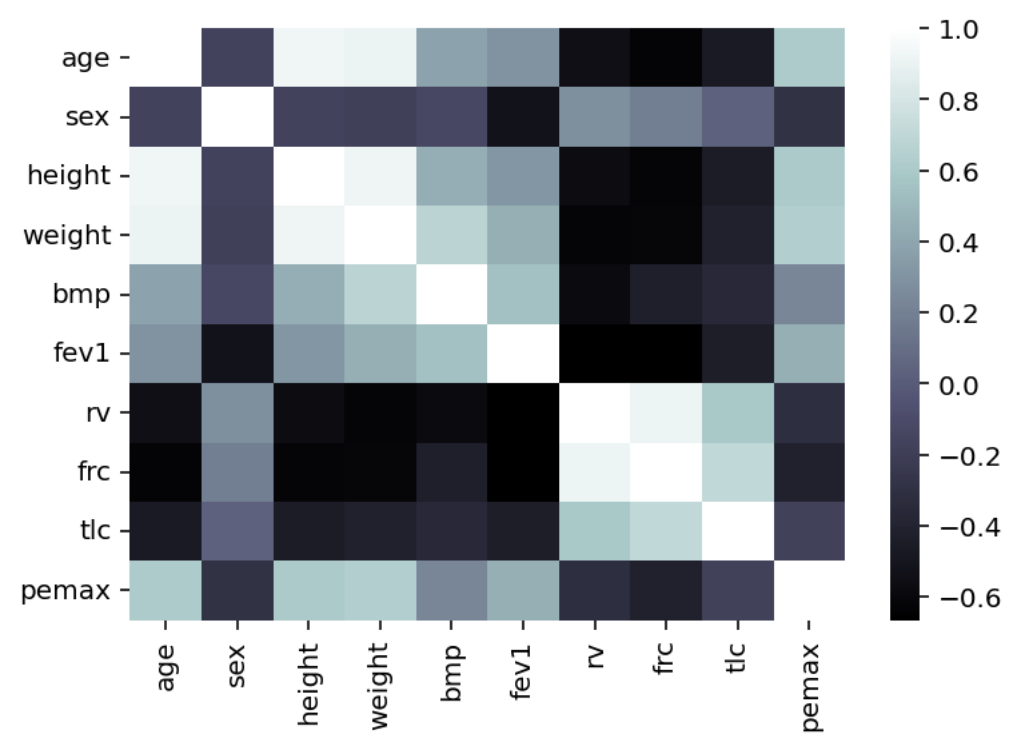
$$cov(x_1, x_2) = cov(x_2, x_1) = E(x_1 x_2) - E(x_1)E(x_2)$$

**covariance**

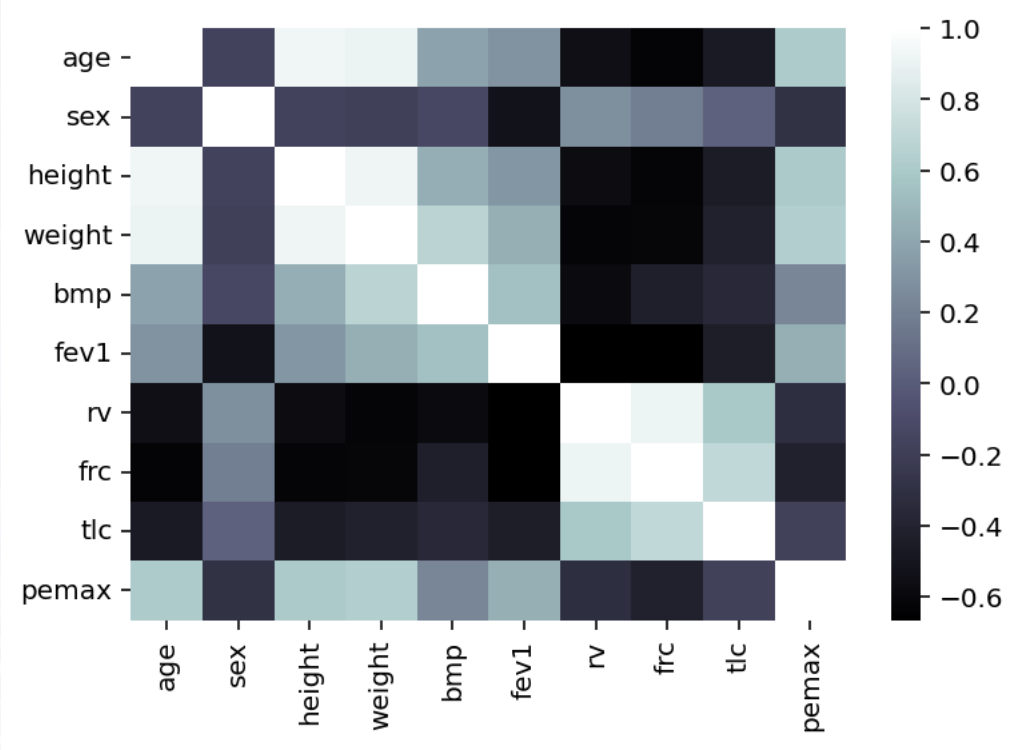
$$\rho(x_1, x_2) = \frac{cov(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

**Pearson's correlation coefficient**

```
sns.heatmap(data.corr(), cmap = "bone")
```




```
sns.heatmap(data_scaled.corr(),
            cmap = "bone")
```

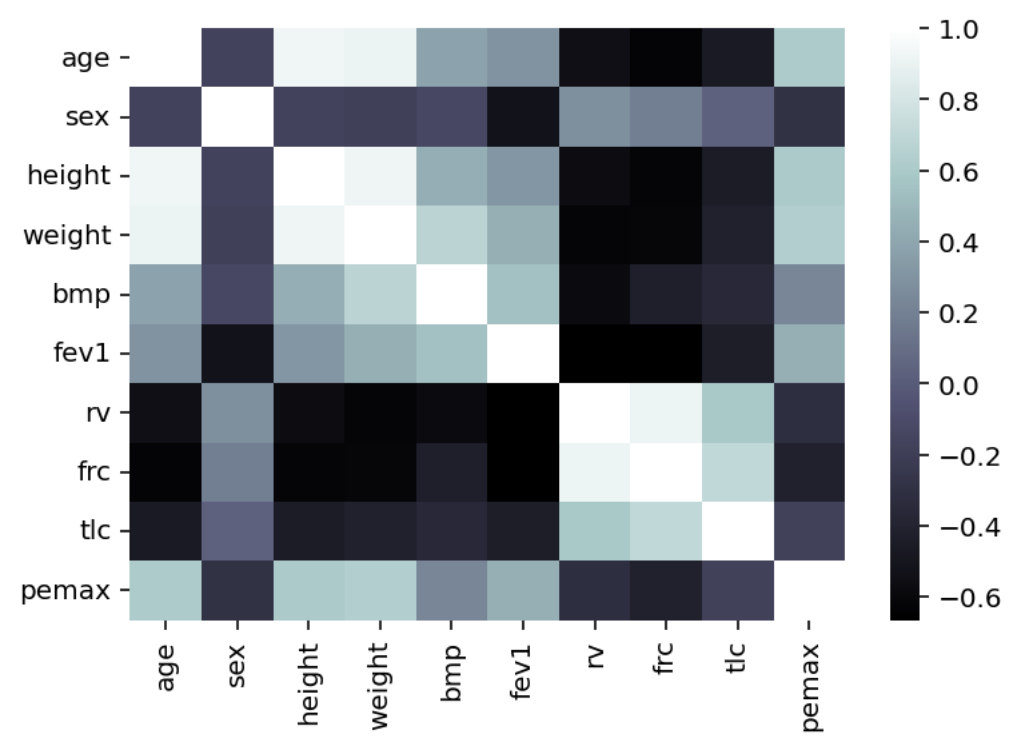


# Correlation, Dimension Reduction and PCA:



correlation **is** scale invariant!

```
sns.heatmap(data.corr(), cmap = "bone")
```

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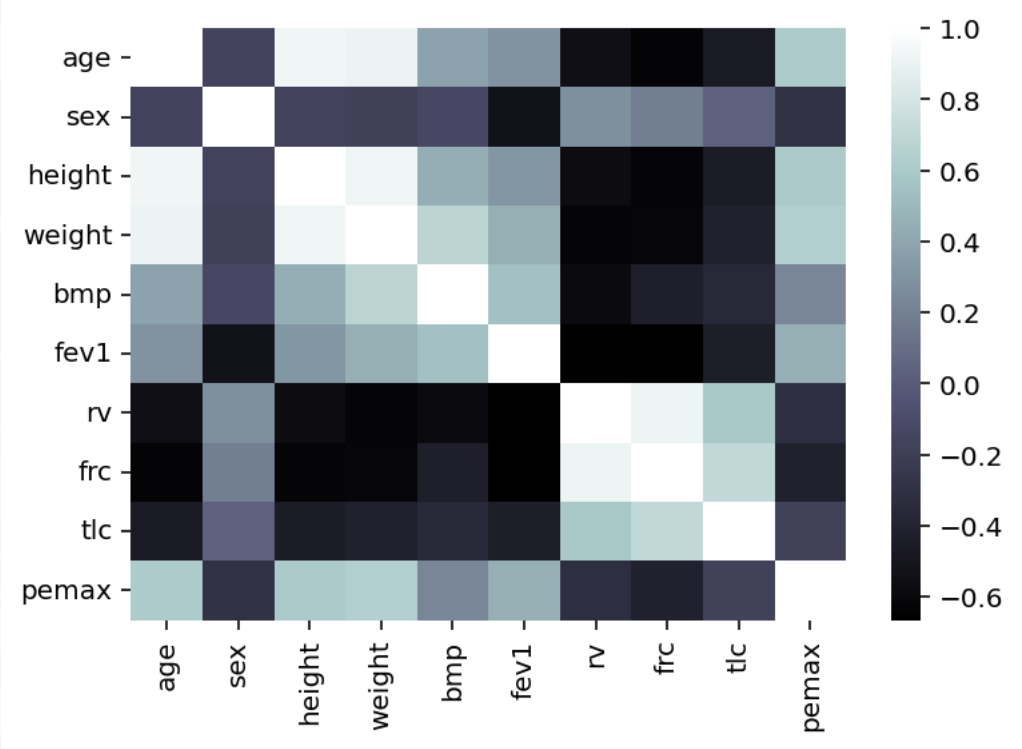


# Correlation, Dimension Reduction and PCA:

$$\rho(x_1, x_2) = \frac{cov(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

$\rho(x_1, x_2)$:

- ranges from -1 to +1
- zero: no correlation **(completely independent)**
- -1: max anti correlation
- +1: max correlation
- **scale invariant**

# Correlation, Dimension Reduction and PCA:



**Important quantities you should know:**

**mean** $$\mu = E(x) = \int x\, p(x)\, dx$$

**variance** $$\sigma^2 = var(x) = \int (x-\mu)^2\, p(x)\, dx$$

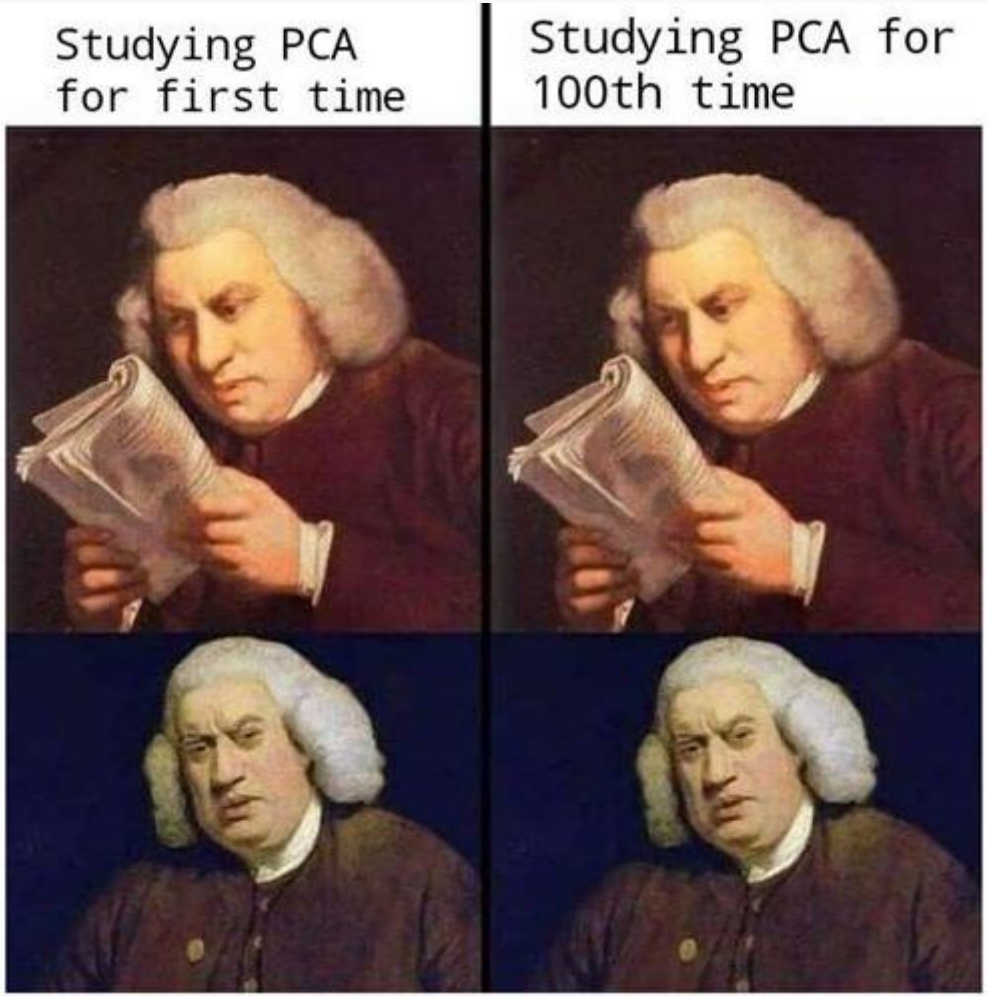
$$\sigma^2 = E(x^2) - E(x)^2$$

$$\sigma_{tot}^2 = \sigma_1^2 + \sigma_2^2 + 2\, cov(x_1, x_2)$$

**covariance** $$cov(x_1, x_2) = E(x_1 x_2) - E(x_1)E(x_2)$$

**correlation coefficient** $$\rho(x_1, x_2) = \frac{cov(x_1, x_2)}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

# Correlation, Dimension Reduction and PCA




credit: Raunak Joshi, @JackRaifer, X

Outline

The curse of dimensionality

Variance and Covariance

Correlation

**Principal Component Analysis (PCA)**

Alternatives

# Correlation, Dimension Reduction and PCA:



**<u>correlation means:</u>**

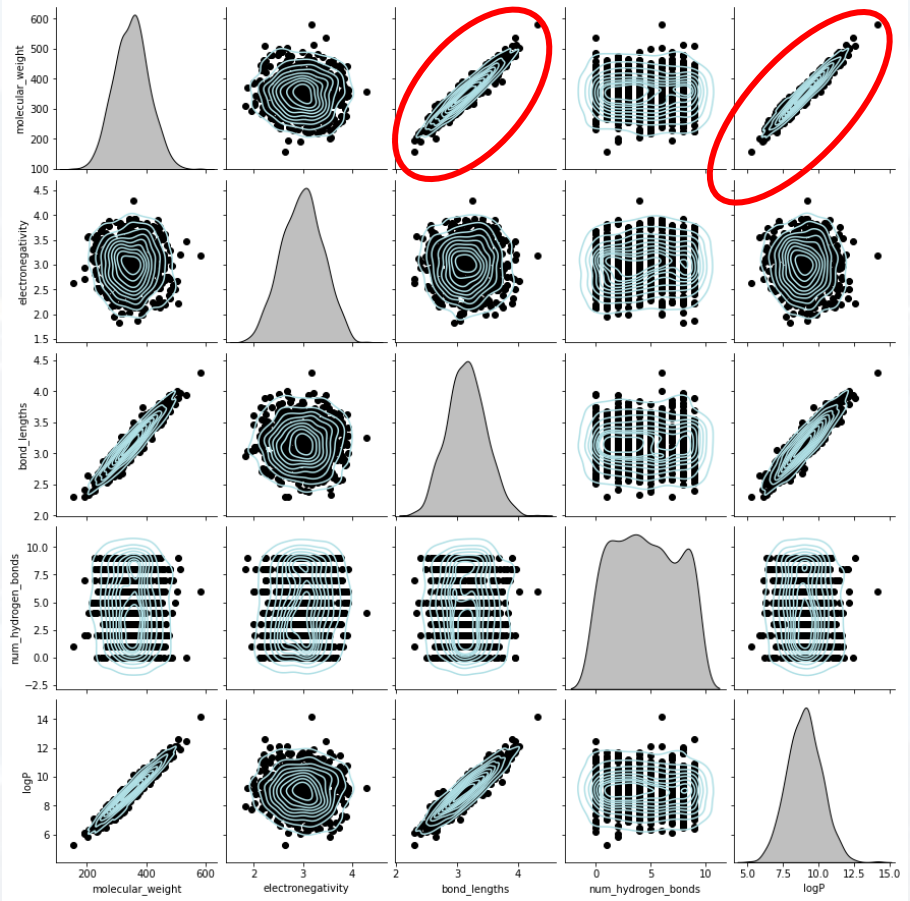
- features are **not mutually independent**

- we can predict feature ***a*** from feature ***b*** to some extend

- we don't need all features

→ **reducing number of features** (dimensions) without losing information

| | label | molecular_weight | electronegativity | bond_lengths | num_hydrogen_bonds | logP |
|---|---|---|---|---|---|---|
| | Toxic | 382.602 | 2.00269 | 3.61153 | 3 | 9.82666 |
| | Toxic | 408.961 | 2.93626 | 3.47904 | 6 | 9.85889 |
| | Non-Toxic | 239.548 | 2.71413 | 2.63922 | 8 | 6.75962 |
| | Non-Toxic | 315.58 | 2.85598 | 2.86034 | 9 | 8.70674 |
| | Non-Toxic | 282.521 | 2.83877 | 2.9664 | 1 | 7.8173 |

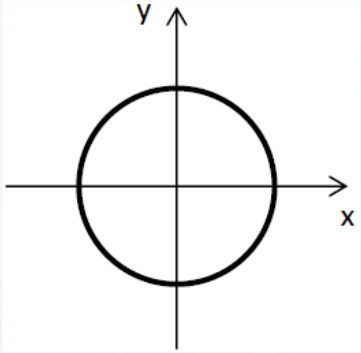


# Correlation, Dimension Reduction and PCA:



## about quadratic forms

**circle**



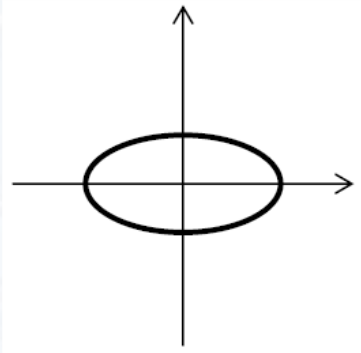

*"distance to a reference point is constant"*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = const$$

$$x^2 + y^2 = const = r^2$$

$$a = b \rightarrow x^2 + y^2 = r^2$$

**ellipse**



*"stretching the coordinate system"*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = const$$

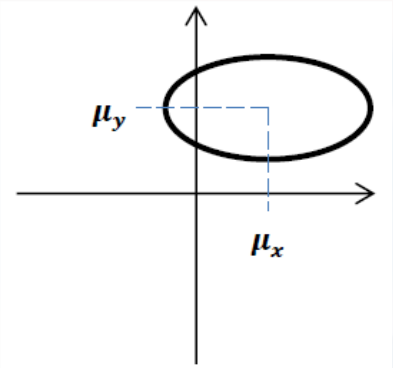
$$a \neq b$$

# Correlation, Dimension Reduction and PCA:



## about quadratic forms

**ellipse**

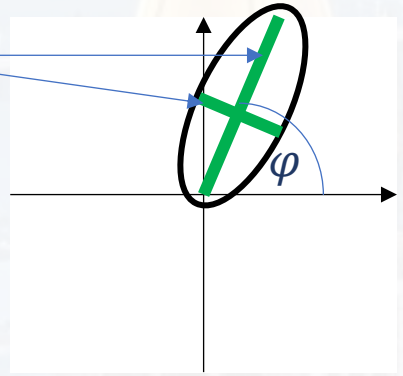



*"moving the center of the ellipse"*

$$\frac{(x-\mu_x)^2}{a^2}+\frac{(y-\mu_y)^2}{b^2}=const$$

$$x_0=0, x_0\rightarrow\mu_x$$
$$y_0=0, y_0\rightarrow\mu_y$$

**principal axes**




*"turning the ellipse by an angle $\varphi$"*

$$\varphi=\frac{1}{2}atan\left(\frac{c}{\frac{1}{a^2}-\frac{1}{b^2}}\right)$$

$$\frac{(x-\mu_x)^2}{a^2}+\frac{(y-\mu_y)^2}{b^2}+c(x-\mu_x)(y-\mu_y)=const$$

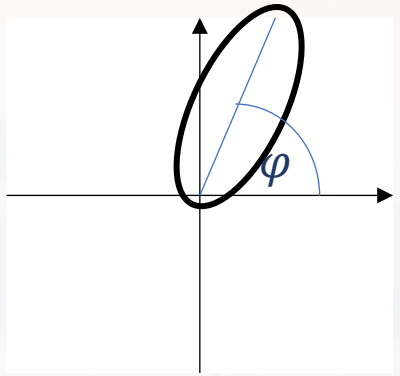
**turning the form**

# Correlation, Dimension Reduction and PCA:



## about quadratic forms

$$\frac{(x-\mu_x)^2}{a^2}+\frac{(y-\mu_y)^2}{b^2}+c(x-\mu_x)(y-\mu_y)=const$$




matrix form:

$$const=\begin{pmatrix}x-\mu_x\\ y-\mu_y\end{pmatrix}^T\begin{pmatrix}1/a^2 & c/2\\ c/2 & 1/b^2\end{pmatrix}\begin{pmatrix}x-\mu_x\\ y-\mu_y\end{pmatrix}$$

often:

$$const=\begin{pmatrix}x-\mu_x\\ y-\mu_y\end{pmatrix}^T\begin{pmatrix}A & C/2\\ C/2 & B\end{pmatrix}\begin{pmatrix}x-\mu_x\\ y-\mu_y\end{pmatrix}$$

more general:

$$const=\begin{pmatrix}x-\mu_x\\ y-\mu_y\\ 1\end{pmatrix}^T\begin{pmatrix}A & C/2 & D/2\\ C/2 & B & E/2\\ D/2 & E/2 & F\end{pmatrix}\begin{pmatrix}x-\mu_x\\ y-\mu_y\\ 1\end{pmatrix}$$

depending on A, B, C, D, E, F

- **circle**
- **ellipse**
- **parabola**
- **hyperbola**

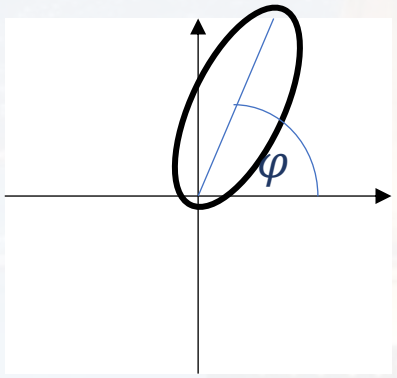
# Correlation, Dimension Reduction and PCA:



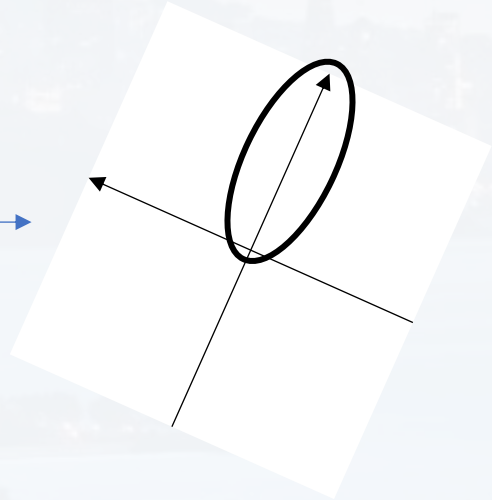
## about quadratic forms

$$\frac{(x-\mu_x)^2}{a^2}+\frac{(y-\mu_y)^2}{b^2}+\boxed{c(x-\mu_x)(y-\mu_y)}=\begin{pmatrix}x-\mu_x\\ y-\mu_y\end{pmatrix}^T\begin{pmatrix}1/a^2 & \boxed{c/2}\\ \boxed{c/2} & 1/b^2\end{pmatrix}\begin{pmatrix}x-\mu_x\\ y-\mu_y\end{pmatrix}=const$$

turning the form




turning the coordinate system



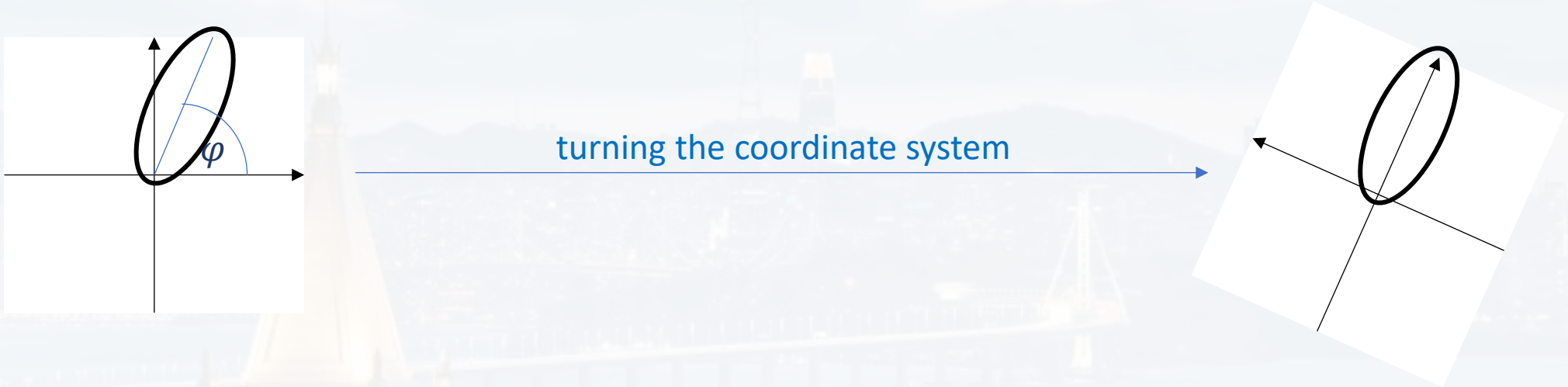
$$\frac{(x_{new}-\mu_{x(new)})^2}{a_{new}^2}+\frac{(y_{new}-\mu_{y(new)})^2}{b_{new}^2}=\begin{pmatrix}x_{new}-\mu_{x(new)}\\ y_{new}-\mu_{y(new)}\end{pmatrix}^T\boxed{\begin{pmatrix}1/a_{new}^2 & 0\\ 0 & 1/b_{new}^2\end{pmatrix}}\begin{pmatrix}x_{new}-\mu_{x(new)}\\ y_{new}-\mu_{y(new)}\end{pmatrix}=const$$

# Correlation, Dimension Reduction and PCA:



## about quadratic forms

non – diagonal elements: - turn/shear the object
diagonal elements: - stretches (or flips, if negative) the object




not turned/sheared → principal axes of the object are **parallel to the coordinate** axes

new coord. axis are called: **eigenvectors** $\vec{v}$ ("eigen", German for "proper")
→ they span the proper coordinate system!

$$\begin{pmatrix} \lambda_1 & 0 \dots & 0 \\ 0 & \lambda_i & 0 \\ 0 & 0 & \lambda_N \end{pmatrix}$$

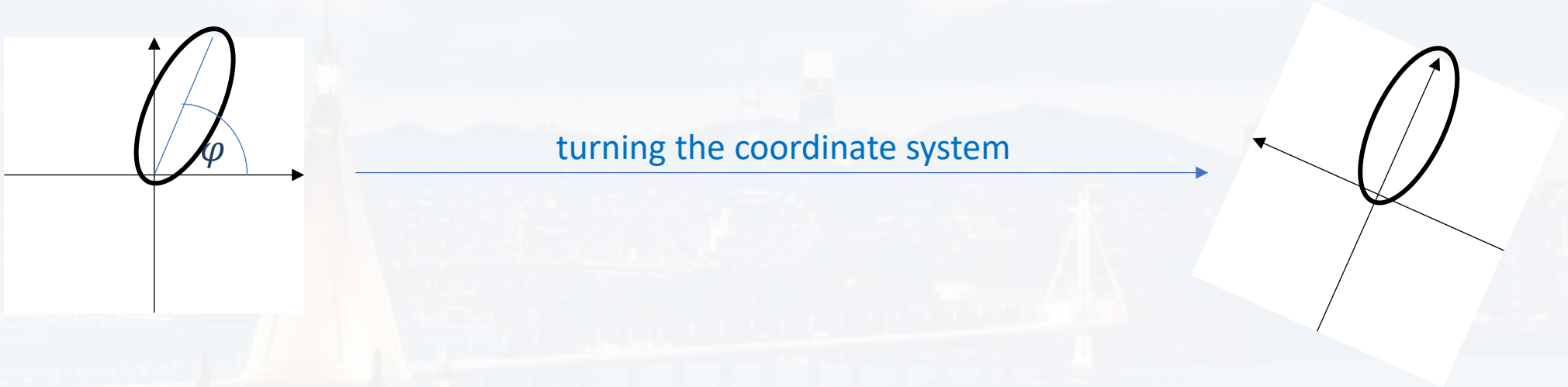
in the proper coordinate system: matrix is diagonal (entries are called **"eigenvalues"** $\lambda$)

# Correlation, Dimension Reduction and PCA:



## about quadratic forms

non – diagonal elements: - turn/shear the object
diagonal elements: - stretches (or flips, if negative) the object




In a coordinate system in which the principal axes are **parallel to the coordinate** axes

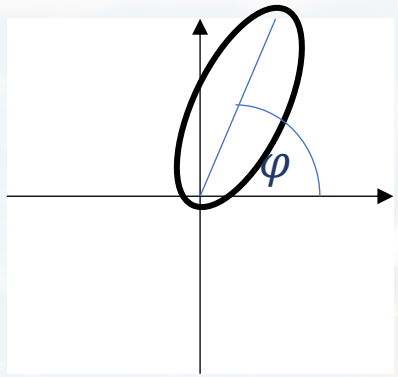
- the matrix that defines the form is **diagonal**
- the **entries** of the now diagonal matrix are called **eigenvalues** $\lambda$
- the **axes** of this coordinates system are called **eigenvectors** $\vec{v}$
- eigen means **"proper"**, i. e. it is the **"most suitable"** coordinate system

$$\begin{pmatrix} \lambda_1 & 0 \dots & 0 \\ 0 & \lambda_i & 0 \\ 0 & 0 & \lambda_N \end{pmatrix}$$

# Correlation, Dimension Reduction and PCA:

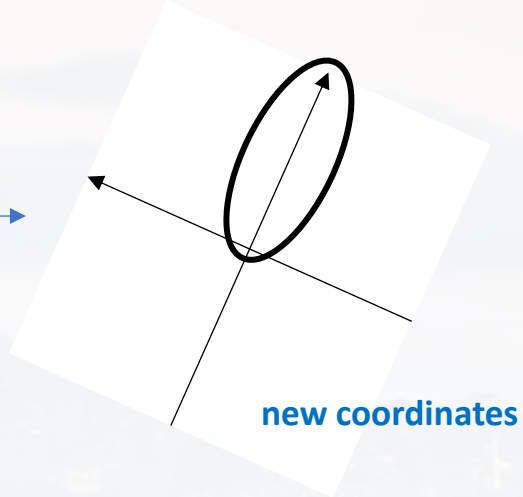


## about quadratic forms




old coordinates

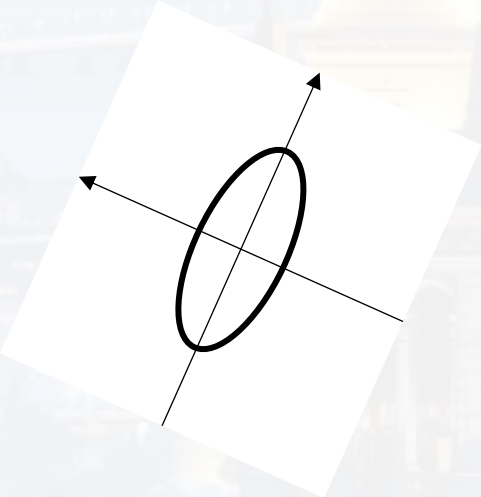
turning the coordinate system

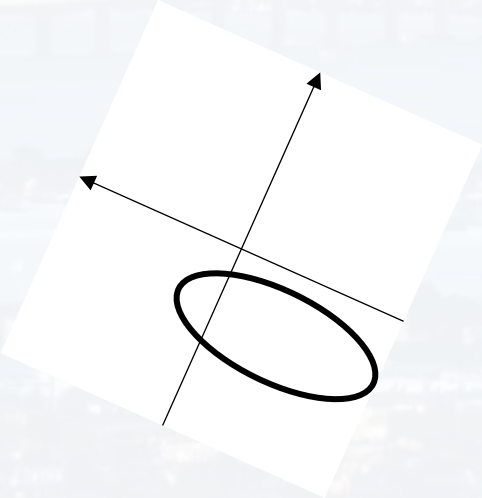


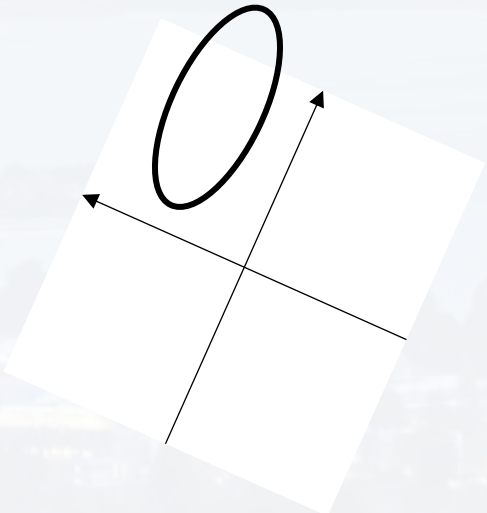
new coordinates

not turned/sheared

→principal axes of the object are **parallel to the coordinate** axis







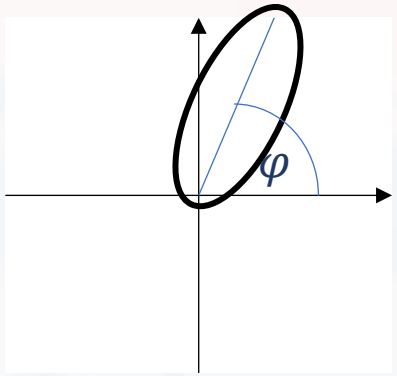
# Correlation, Dimension Reduction and PCA:



…so, what PCA actually is:

1) turned ellipse

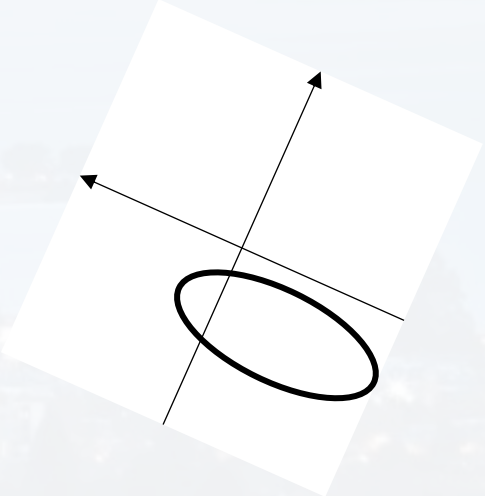
$$\begin{pmatrix} x \\ y \end{pmatrix}^T \underbrace{\begin{pmatrix} A & C/2 \\ C/2 & B \end{pmatrix}}_{M} \begin{pmatrix} x \\ y \end{pmatrix} = A\,x^2 + B\,y^2 + C\,xy$$




2) turning the coordinate system, such that principal axes of the are **parallel to the coordinate**

$$\begin{pmatrix} x_{new} \\ y_{new} \end{pmatrix}^T \underbrace{\begin{pmatrix} \boldsymbol{\lambda}_1 & 0 \\ 0 & \boldsymbol{\lambda}_2 \end{pmatrix}}_{M_{new}} \begin{pmatrix} x_{new} \\ y_{new} \end{pmatrix} = \boldsymbol{\lambda}_1 {x_{new}}^2 + \boldsymbol{\lambda}_2 {y_{new}}^2$$

eigenvalues $\boldsymbol{\lambda}$



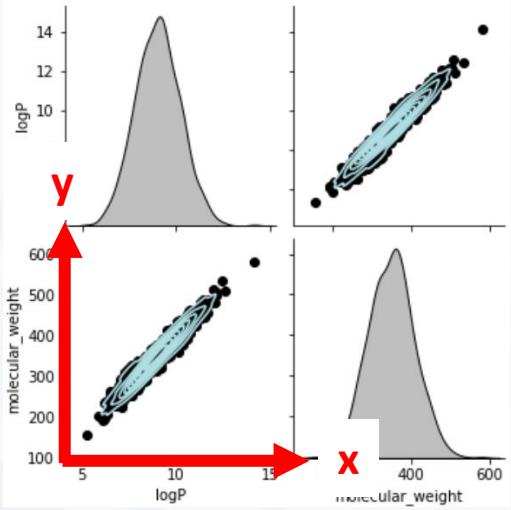
But what does it have to do with covariance and correlation?

# Correlation, Dimension Reduction and PCA:

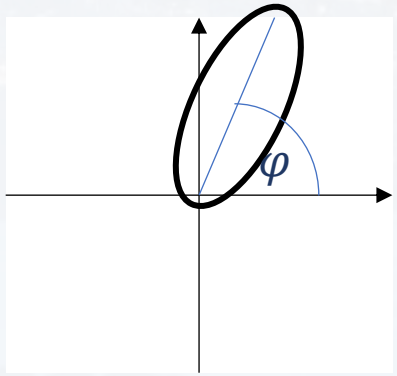


$$\sigma_{tot}^2 = \boxed{\sigma_x^2} + \boxed{\sigma_y^2} + \boxed{2\,cov(x,y)}$$

$$= \boxed{\sum_i^N (x_i - \mu_x)^2} + \boxed{\sum_j^M (y_j - \mu_y)^2} + \boxed{2\sum_j^M \sum_i^N (x_i - \mu_x)(y_j - \mu_y)}$$




$$const = \boxed{\frac{(x-\mu_x)^2}{a^2}} + \boxed{\frac{(y-\mu_y)^2}{b^2}} + \boxed{2\,c(x-\mu_x)(y-\mu_y)}$$




**It is the same structure!**

$$const = \begin{pmatrix} \boldsymbol{x} - \boldsymbol{\mu_x} \\ \boldsymbol{y} - \boldsymbol{\mu_y} \end{pmatrix}^T \begin{pmatrix} 1/a^2 & c \\ c & 1/b^2 \end{pmatrix} \begin{pmatrix} \boldsymbol{x} - \boldsymbol{\mu_x} \\ \boldsymbol{y} - \boldsymbol{\mu_y} \end{pmatrix}$$

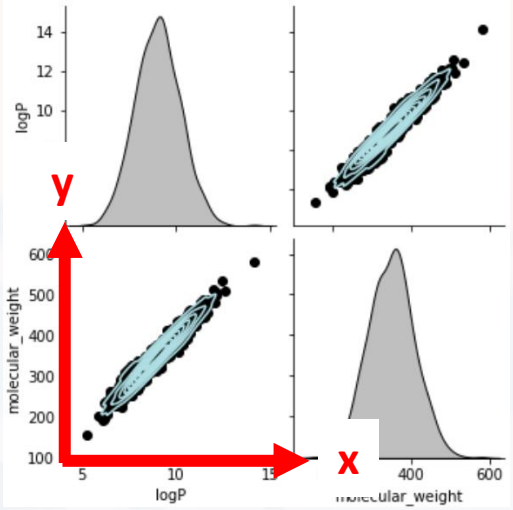
$$C = \begin{pmatrix} \boldsymbol{x} - \boldsymbol{\mu_x} \\ \boldsymbol{y} - \boldsymbol{\mu_y} \end{pmatrix}^T \begin{pmatrix} \sigma_x^2 & cov(y,x) \\ cov(x,y) & \sigma_y^2 \end{pmatrix} \begin{pmatrix} \boldsymbol{x} - \boldsymbol{\mu_x} \\ \boldsymbol{y} - \boldsymbol{\mu_y} \end{pmatrix} \qquad cov(y,x) = cov(x,y)$$

# Correlation, Dimension Reduction and PCA:

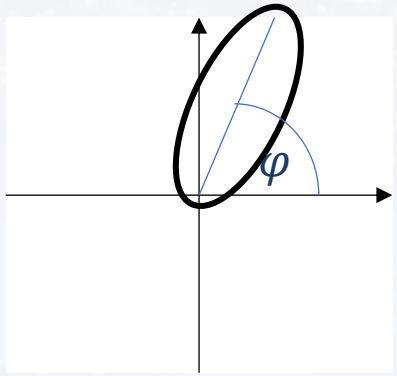


$$\sigma_{tot}^2 = \sigma_x^2 + \sigma_y^2 + \boxed{2\,cov(x,y)}$$

$$= \sum_i^N (x_i - \mu_x)^2 + \sum_j^M (y_j - \mu_y)^2 + \boxed{2\sum_j^M \sum_i^N (x_i - \mu_x)(y_j - \mu_y)}$$




$$const = \frac{(x-\mu_x)^2}{a^2} + \frac{(y-\mu_y)^2}{b^2} + \boxed{2\,c(x-\mu_x)(y-\mu_y)}$$




**It is the same structure!**

$$const = \begin{pmatrix} \boldsymbol{x}-\boldsymbol{\mu_x} \\ \boldsymbol{y}-\boldsymbol{\mu_y} \end{pmatrix}^T \begin{pmatrix} 1/a^2 & \boxed{c} \\ \boxed{c} & 1/b^2 \end{pmatrix} \begin{pmatrix} \boldsymbol{x}-\boldsymbol{\mu_x} \\ \boldsymbol{y}-\boldsymbol{\mu_y} \end{pmatrix}$$

$$C = \begin{pmatrix} \boldsymbol{x}-\boldsymbol{\mu_x} \\ \boldsymbol{y}-\boldsymbol{\mu_y} \end{pmatrix}^T \begin{pmatrix} \sigma_x^2 & \boxed{cov(y,x)} \\ \boxed{cov(x,y)} & \sigma_y^2 \end{pmatrix} \begin{pmatrix} \boldsymbol{x}-\boldsymbol{\mu_x} \\ \boldsymbol{y}-\boldsymbol{\mu_y} \end{pmatrix} \qquad cov(y,x) = cov(x,y)$$

# Correlation, Dimension Reduction and PCA:



$$C = \begin{pmatrix} \boldsymbol{x} - \boldsymbol{\mu_x} \\ \boldsymbol{y} - \boldsymbol{\mu_y} \end{pmatrix}^T \underbrace{\begin{pmatrix} \sigma_x^2 & \boxed{cov(y,x)} \\ \boxed{cov(x,y)} & \sigma_y^2 \end{pmatrix}}_{\textit{covariance matrix } \Sigma} \begin{pmatrix} \boldsymbol{x} - \boldsymbol{\mu_x} \\ \boldsymbol{y} - \boldsymbol{\mu_y} \end{pmatrix} \qquad cov(y,x) = cov(x,y)$$

- geometrically, the **covariance matrix** can be interpreted as quadratic form
- the covariances are the **non-diagonal** elements of the **covariance matrix**
- aim: finding a coordinate transformation, where the **covariance matrix** is diagonal

$$\begin{pmatrix} \lambda_1 \ddots & 0 \dots & 0 \\ 0 & \lambda_i \ddots & 0 \\ 0 & 0 & \lambda_I \end{pmatrix}$$

**the diagonal entries are called eigenvalues** (= variances in new coordinate system)

→ all variables **are independent**

→ principal components of the **covariance matrix** are **parallel** to the **new coordinate axes** **(= eigenvectors)**

# Correlation, Dimension Reduction and PCA:



How to interpret the eigenvalues?

$$\begin{pmatrix} \boldsymbol{\lambda_1} & 0 \dots & 0 \\ 0 & \boldsymbol{\lambda_i} & 0 \\ 0 & 0 & \boldsymbol{\lambda_I} \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \sigma_{11}^2 & \dots cov(x_1, x_j) \dots & cov(x_1, x_I) \\ \dots & \dots & \dots \\ cov(x_i, x_1) & \dots \sigma_{ii}^2 \dots & cov(x_i, x_I) \\ \dots & \dots & \dots \\ cov(x_I, x_1) & \dots cov(x_I, x_j) \dots & \sigma_{II}^2 \end{pmatrix}$$

**PCA**

$$\Sigma_{new} = \begin{pmatrix} \boldsymbol{\sigma_{11}^2(new)} & \dots 0 \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots \boldsymbol{\sigma_{ii}^2(new)} \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots 0 \dots & \boldsymbol{\sigma_{II}^2(new)} \end{pmatrix}$$

# Correlation, Dimension Reduction and PCA:



How to interpret the eigenvalues?

$$\begin{pmatrix} \boldsymbol{\lambda_1} & 0 \dots & 0 \\ 0 & \boldsymbol{\lambda_i} & 0 \\ 0 & 0 & \boldsymbol{\lambda_I} \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \sigma_{11}^2 & \dots cov(x_1, x_j) \dots & cov(x_1, x_I) \\ \dots & \dots & \dots \\ cov(x_i, x_1) & \dots \sigma_{ii}^2 \dots & cov(x_i, x_I) \\ \dots & \dots & \dots \\ cov(x_I, x_1) & \dots cov(x_I, x_j) \dots & \sigma_{II}^2 \end{pmatrix}$$

**PCA**

$$\Sigma_{new} = \begin{pmatrix} \boldsymbol{\sigma_{11}^2(new)} & \dots 0 \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots \boldsymbol{\sigma_{ii}^2(new)} \dots & 0 \\ \dots & \dots & \dots \\ 0 & \dots 0 \dots & \boldsymbol{\sigma_{II}^2(new)} \end{pmatrix}$$

the **eigenvalues** of the **covariance matrix** are the **variances** in the **new coordinate** system in which the **principal components** of the **covariance matrix** are **parallel** to the new coordinate axis (= **eigenvectors**)

# Correlation, Dimension Reduction and PCA:

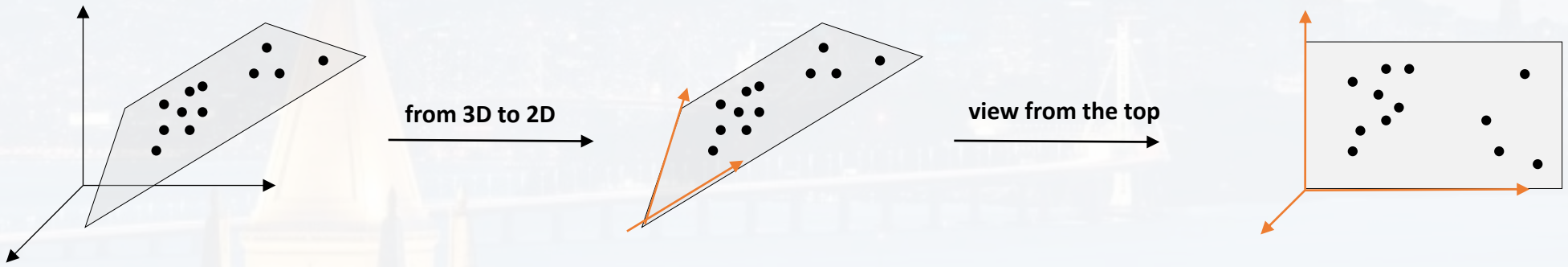


How to interpret the eigenvalues?

$$\begin{pmatrix} \lambda_1 & 0 \dots & 0 \\ 0 & \lambda_i & 0 \\ 0 & 0 & \lambda_I \end{pmatrix}$$

the **eigenvalues** of the **covariance matrix** are the **variances** in the **new coordinate** system

What if eigenvalues are close to zero?




each data point is represented by **three** features...

... but those **features correlate** (x, y) → z

**new coordinate** system

# Correlation, Dimension Reduction and PCA:

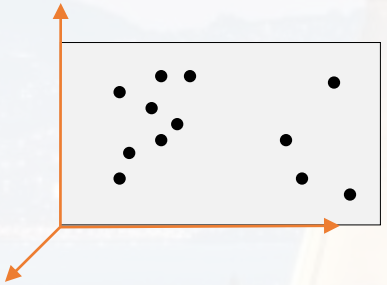


How to interpret the eigenvalues?

$$\begin{pmatrix} \lambda_1 & 0 \dots & 0 \\ 0 & \lambda_i & 0 \\ 0 & 0 & \lambda_I \end{pmatrix}$$

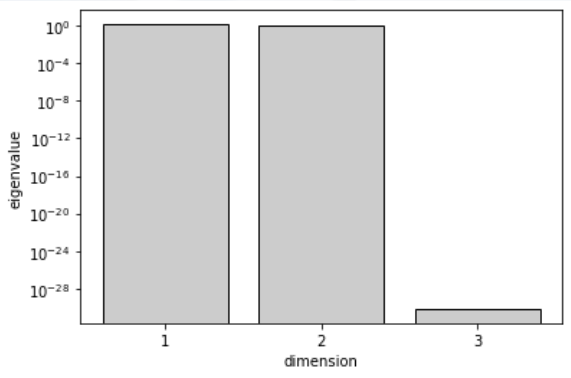
the **eigenvalues** of the **covariance matrix** are the **variances** in the **new coordinate** system

What if eigenvalues are close to zero?



some variance in $x_{new}$, $y_{new}$

almost no variance in $z_{new}$
→ because data exists on 2D, not 3D
→ dimensionality reduction!!




**new coordinate** system

We can reduce the complexity of the data set without loosing information

# Correlation, Dimension Reduction and PCA:



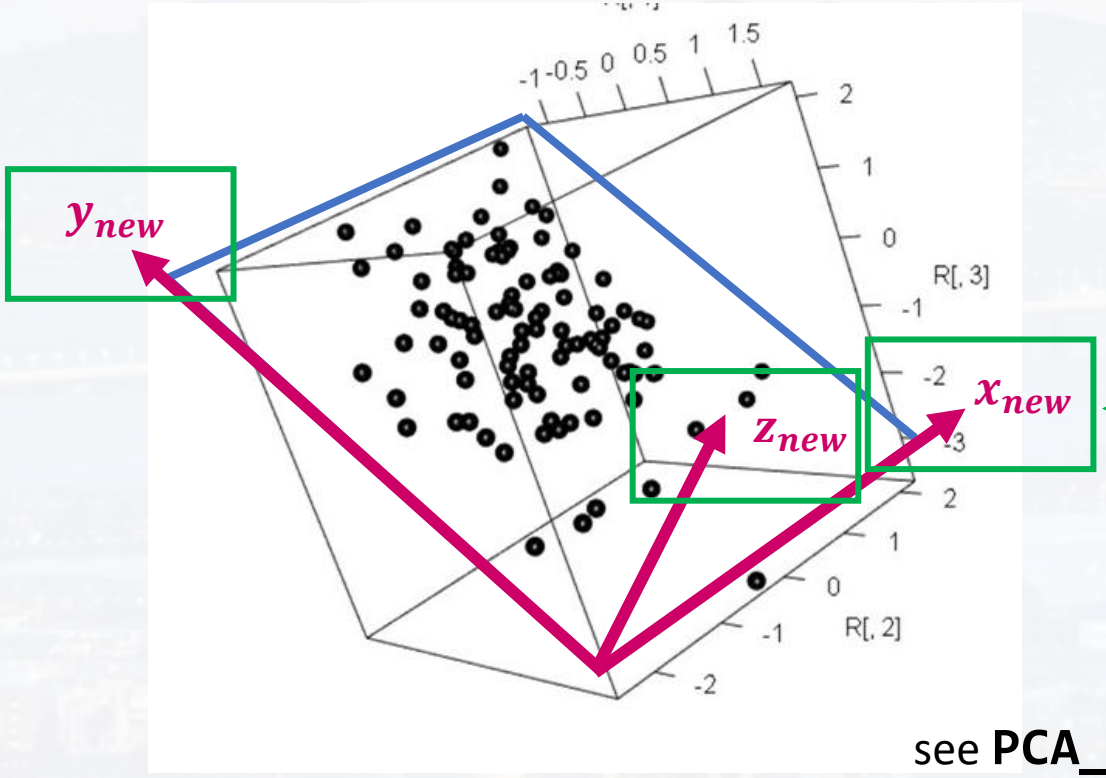
How to interpret the eigenvalues?

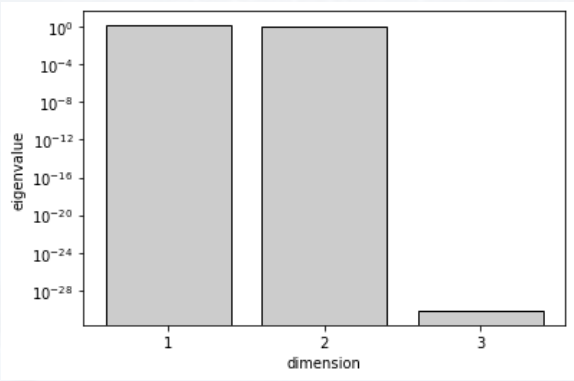
$$\begin{pmatrix} \lambda_1 & 0 \dots & 0 \\ 0 & \lambda_i & 0 \\ 0 & 0 & \lambda_I \end{pmatrix}$$

the **eigenvalues** of the **covariance matrix** are the **variances** in the **new coordinate** system

What if eigenvalues are close to zero?

each eigenvector is a **linear combination** of the old coordinate axes




see **PCA_simple.ipynb**

We can reduce the complexity of the data set without loosing information

# Correlation, Dimension Reduction and PCA:



**Summary:**

- PCA transforms the data into a proper ( = **eigen**) coordinate system
- The coordinate axis of this coordinate system are called **eigenvectors**
- Covariance terms in the covariance matrix $\Sigma_{new}$ disappear
- Therefore, $\boldsymbol{\Sigma_{new}}$ **is a diagonal matrix**
- The diagonal elements are called **eigenvalues**…
- …which are the **variances of the data in the proper (=eigen) coordinate system**
- If some eigenvalues are smaller than others, it means there is less variance in that direction
- We only select those directions **with large eigenvalues** (90%, 95% of total variance)
- and therefore **reduce dimensionality**

# Correlation, Dimension Reduction and PCA:



**Note:**

- PCA **is note scale invariant** because it acts on variances/covariances which are **not scale invariant**

- **scale/normalize data first!**

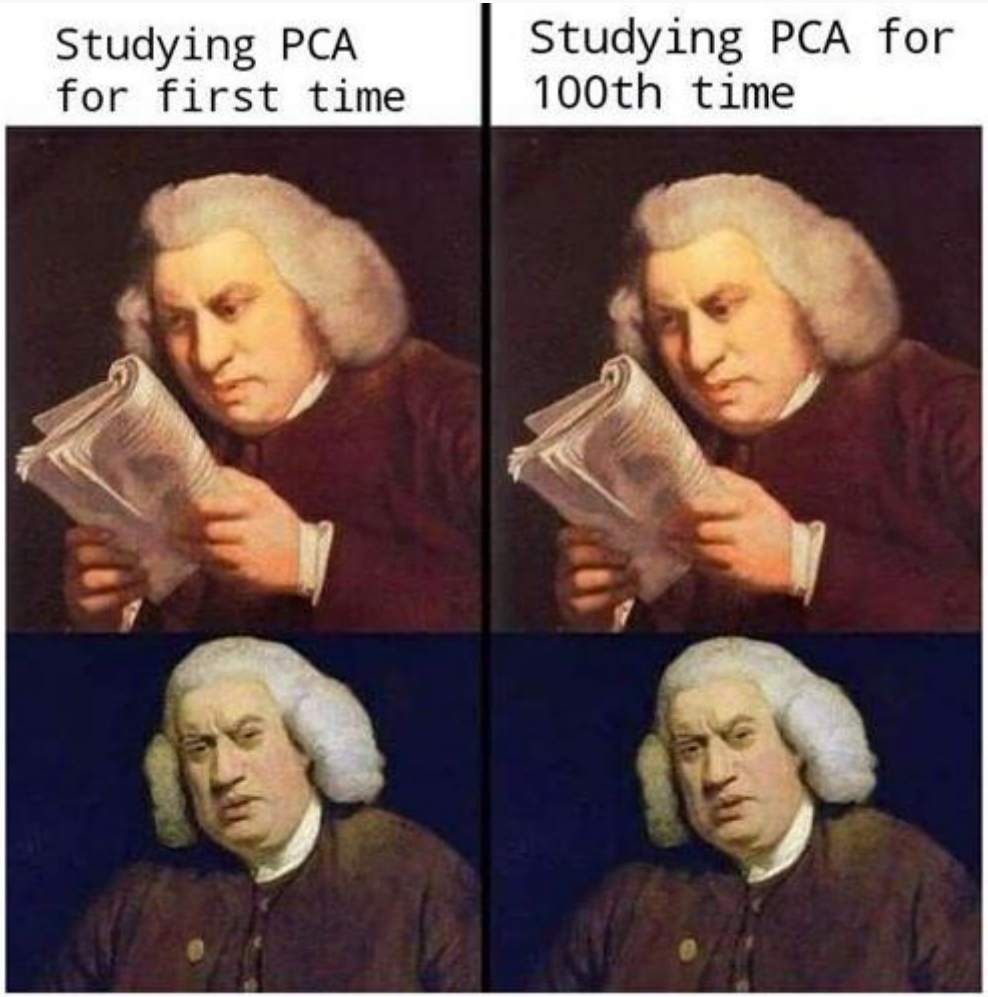
- run PCA before you continue with any ML method

- PCA is a linear operation ( = linear algebra). It does not work if data lives on a curved space (manifold)
  → flat eigenvalue spectrum could be an evidence that PCA is not applicable

- Alternatives are SVMs, LDA, and UMAP for visualization

It is all still a bit abstract, so let us explore **`PCA_Molecules.ipynb`**

# Correlation, Dimension Reduction and PCA




credit: Raunak Joshi, @JackRaifer, X

Outline

The curse of dimensionality

Variance and Covariance

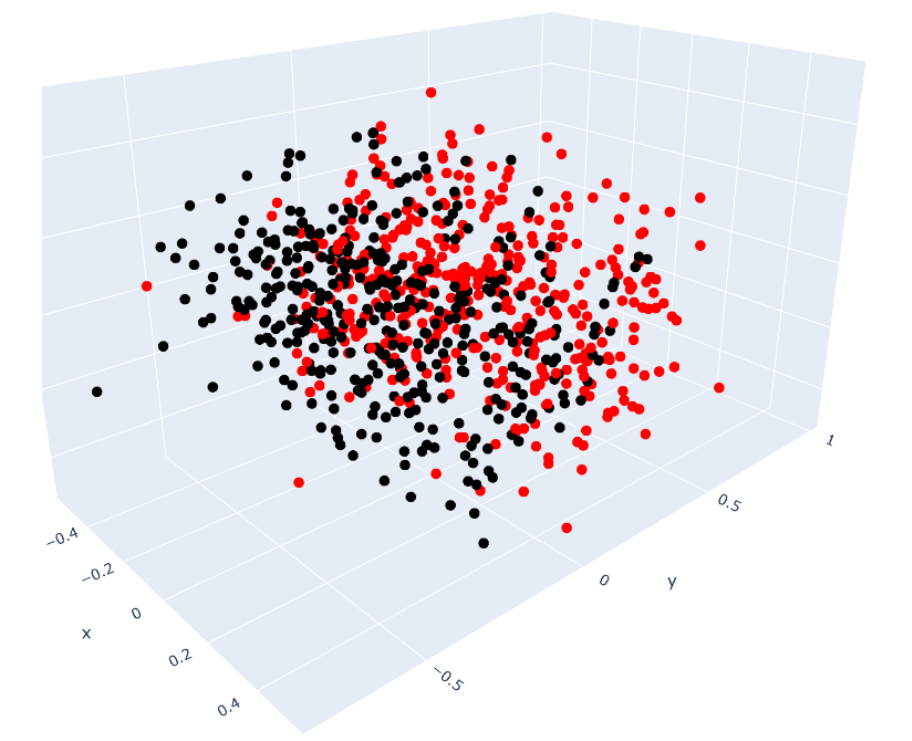
Correlation

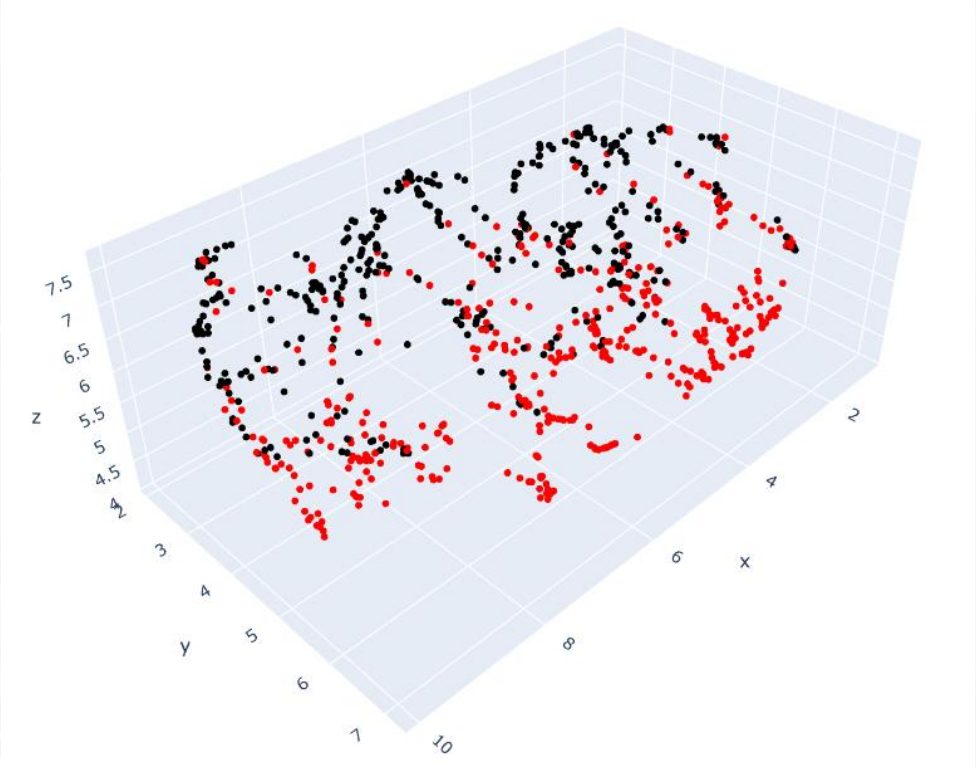
Principal Component Analysis (PCA)

**Alternatives**

# Correlation, Dimension Reduction and PCA: Alternatives

Since PCA reduces dimensionality → also a **visualization tool** by plotting the three most dominant directions:

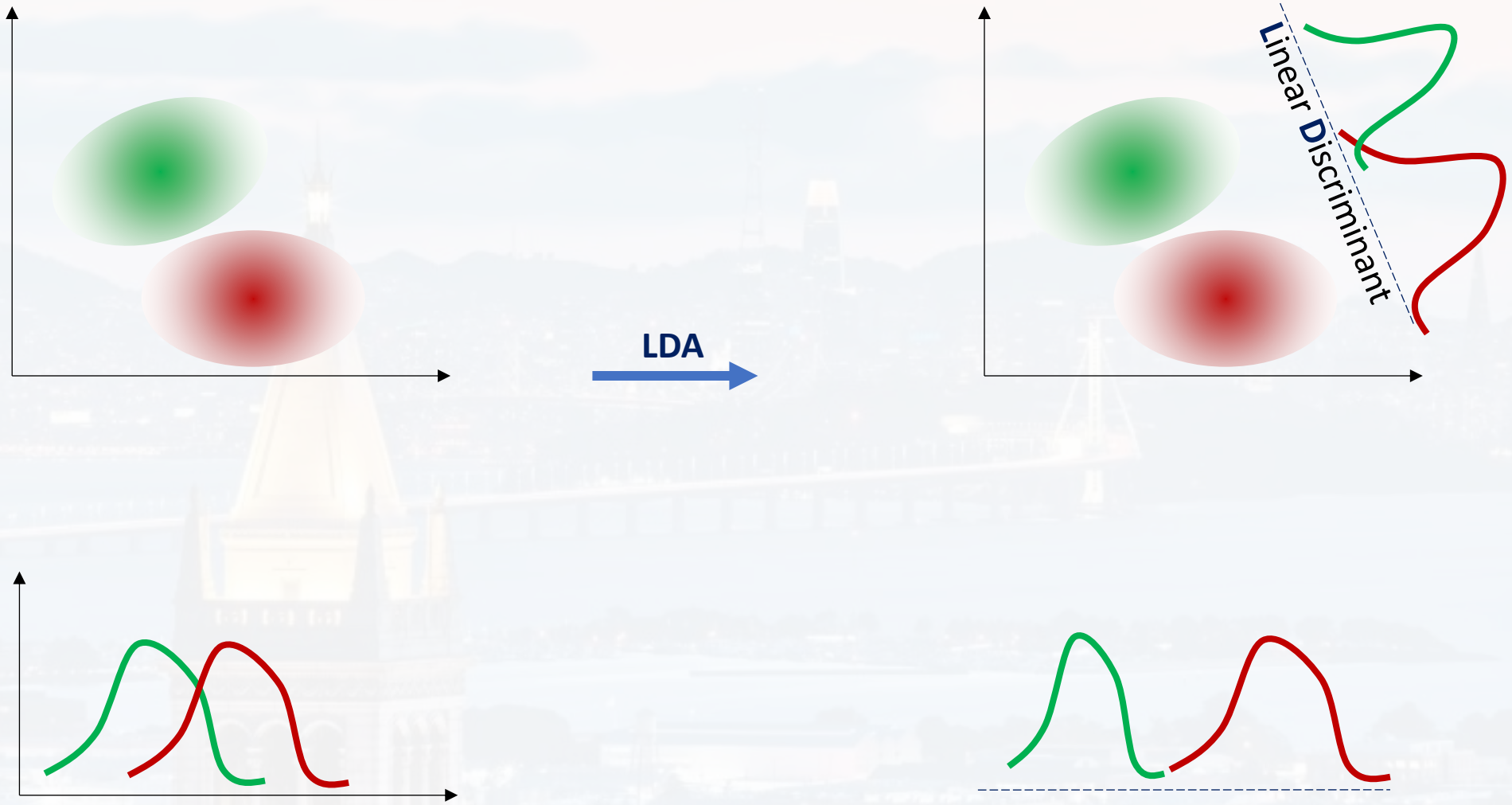



Another common **visualization tool** is UMAP (see next module) that maps n-dimensional data to 2D or 3D

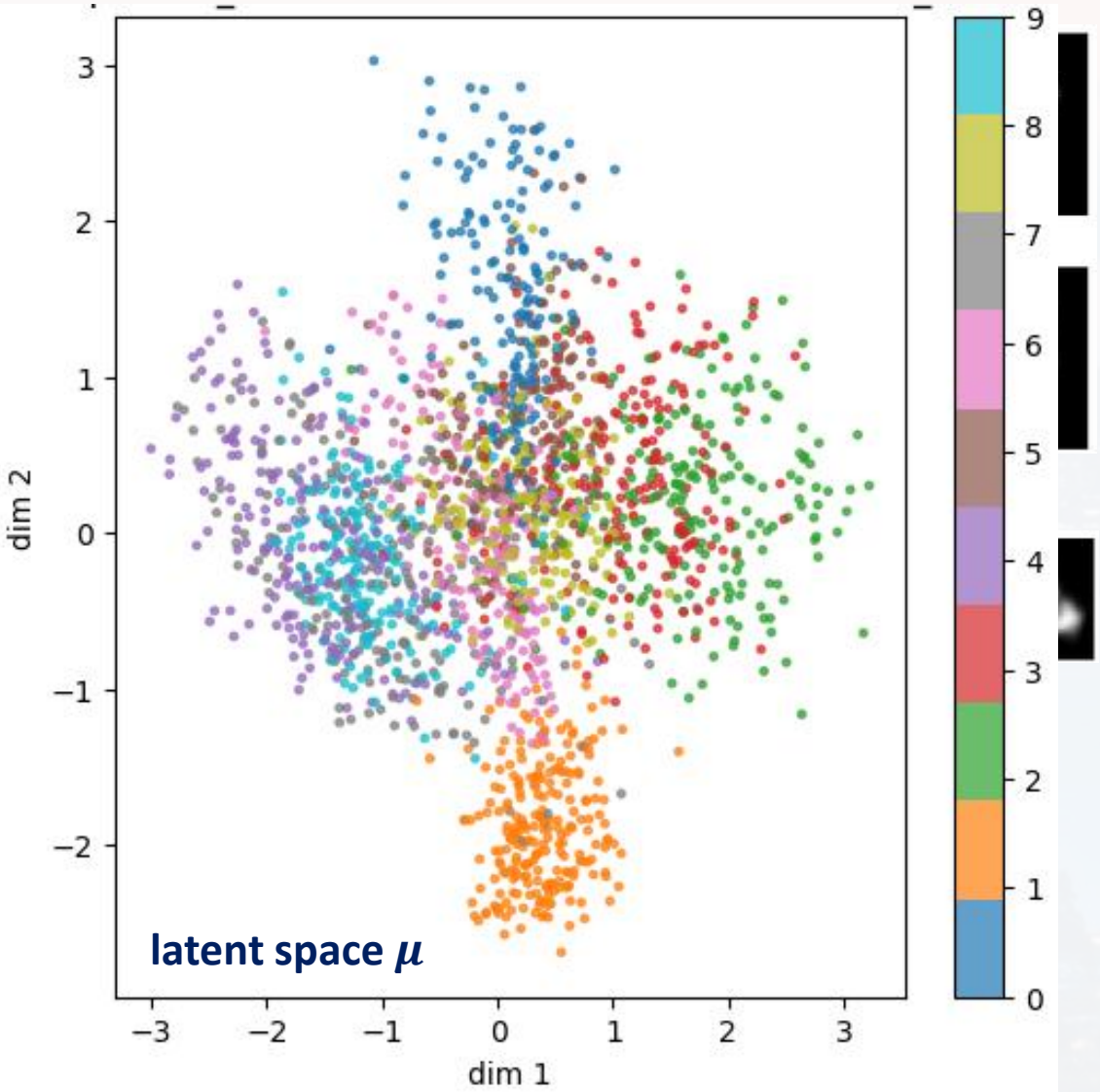
# Correlation, Dimension Reduction and PCA: Alternatives

**L**inear **D**iscriminant **A**nalysis:



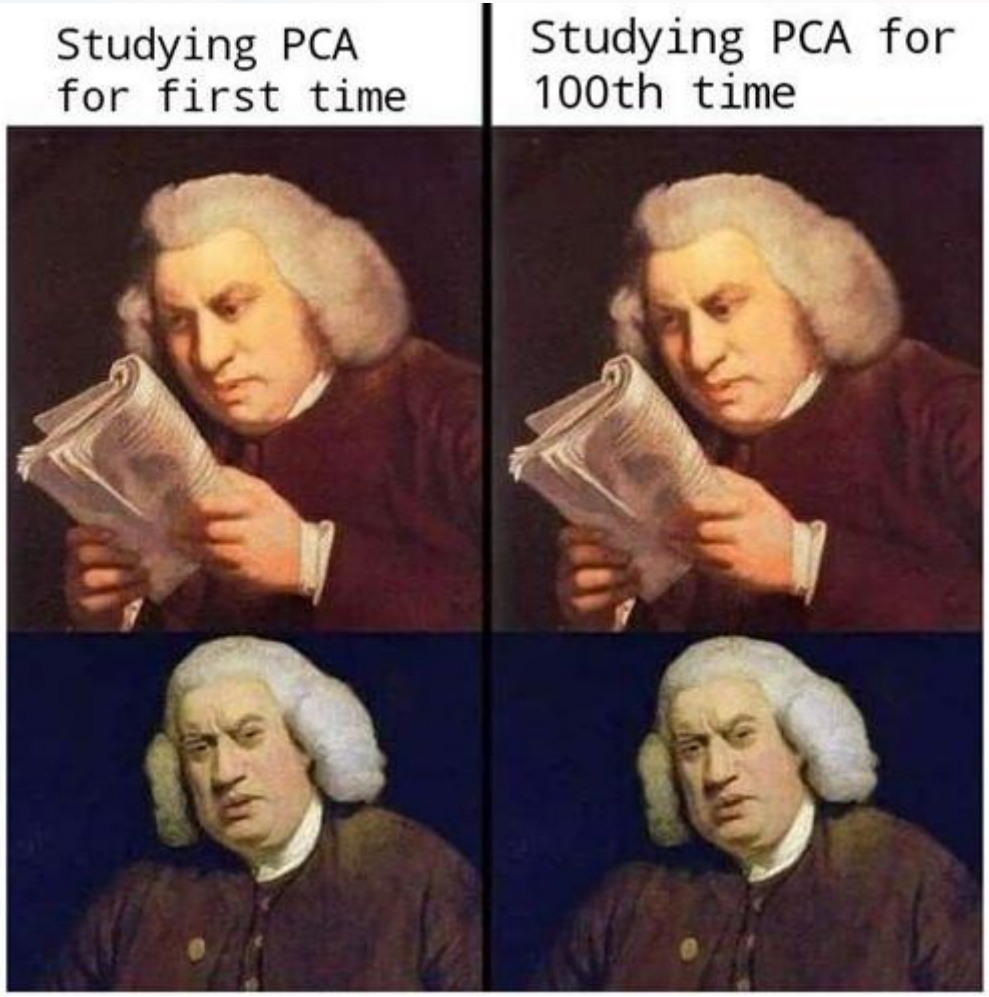
# Correlation, Dimension Reduction and PCA: Alternatives

**(Variational) Autoencoders** (Chem 277B)
an ANN that learns how to map data into a low dimensional "latent space"



# Correlation, Dimension Reduction and PCA

Thank you very much for your attention!




credit: Raunak Joshi, @JackRaifer, X